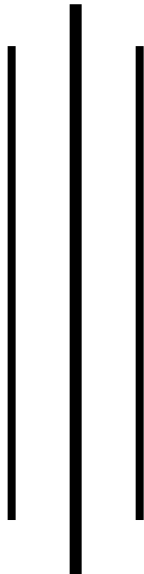


**INSTITUTE OF ENGINEERING
PULCHOWK CAMPUS
DEPARTMENT OF MECHANICAL
ENGINEERING**



**FUNDAMENTALS OF
THERMODYNAMICS AND HEAT
TRANSFER**

[TUTORIAL SHEETS]

ELECTRICAL, ELECTRONICS AND COMPUTER (I/II)

2071

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TUTORIAL NO: 1

THERMODYNAMIC PROPERTIES

1. A container has two compartments separated by a membrane as shown in **Figure P1.1**. Compartment A has 2 kg of air and a volume of 1 m³; compartment B has 1.5 m³ of air with a specific volume of 2.5 m³/kg. If the membrane is broken, determine the resultant specific volume.
2. A container having two compartments contains steam as shown in **Figure P1.2**. The specific volume of steam in compartment B is 5 m³/kg. The membrane breaks and the resulting specific volume is 8 m³/kg. Find the original specific volume of steam in compartment A.
3. A cylinder with a total volume of 1 m³ has a movable piston as shown in **Figure P1.3**. When the piston is at one fourth of the length, both sides have same specific volume of 4 m³/kg. Determine the specific volumes of both sides when the piston is at middle of the cylinder.

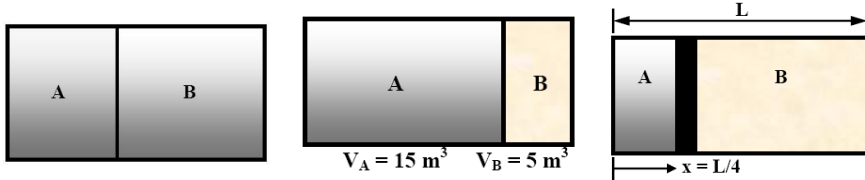


Figure P1.1

Figure P1.2

Figure P1.3

4. An oxygen cylinder having a volume of 10 m³ initially contains 5 kg of oxygen. Determine the specific volume of oxygen in the cylinder initially. During certain process 3 kg of oxygen is consumed, determine the final specific volume of oxygen in the cylinder. Also plot the amount of oxygen that has been consumed versus the specific volume of the remaining in the cylinder.
5. **Figure P1.5** shows a tank within a tank, each containing air. Pressure gage A is located inside the tank B and reads 140 kPa. The U-tube manometer connected to tank B contains mercury. Using data on the diagram, determine the absolute pressures inside the tank A and tank B.
6. A large chamber is separated into two compartments which are maintained different pressures, as shown **Figure P1.6**. Pressure gauge A reads 180 kPa, and pressure gauge B reads 120 kPa. If the barometric pressure is 100 kPa, determine the absolute pressure existing in the compartments and the reading of gauge C.

7. A piston cylinder arrangement shown in **Figure P1.7** has a cross sectional area of 0.01 m² and a piston mass of 80 kg. If atmospheric pressure is 100 kPa, what should be the gas pressure to lift the piston? [Take $g = 9.81 \text{ m/s}^2$]
8. For the piston cylinder device shown in **Figure P1.8**, determine the absolute pressure inside the device. [Take $P_{atm} = 101.3 \text{ kPa}$]

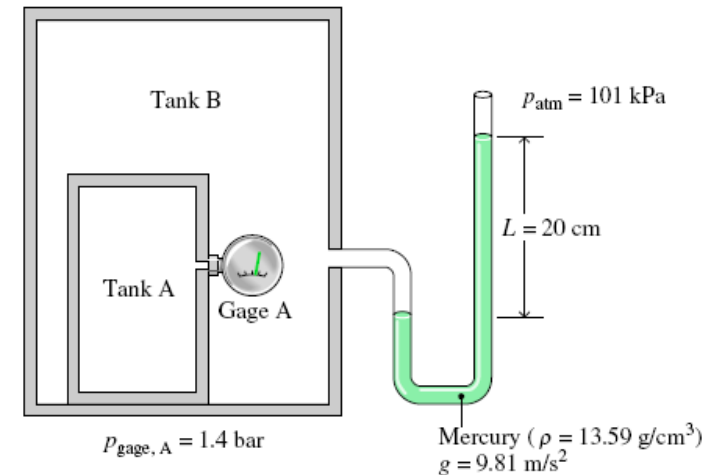


Figure P1.5

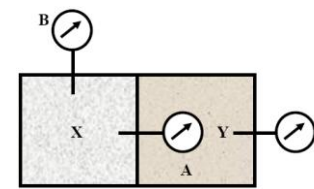


Figure P1.6



Figure P1.7

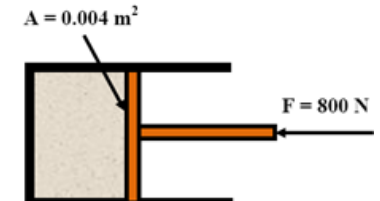


Figure P1.8

9. A gas is contained in a vertical, frictionless piston-cylinder device shown in **Figure P1.9**. The piston has a mass of 4 kg and a cross-sectional area of 35 cm². A compressed spring above the piston exerts a force of 60 N on the piston. If the atmospheric pressure is 95 kPa, determine the pressure inside the cylinder.
10. A 15 kg piston in a cylinder with diameter of 0.15 m is loaded with a linear spring and the outside atmospheric pressure of 100 kPa, as shown in **Figure P1.10**. The spring exerts no force on the piston when it is at the bottom of the cylinder and for the state shown, the pressure is 300 kPa with volume of 0.02

m^3 . The valve is opened to let some air in, causing the piston to rise 5 cm. Find the new pressure. [Take $g = 9.81 \text{ m/s}^2$]

11. Air contained within a vertical piston–cylinder assembly is shown in **Figure P1.11**. On its top, the 10-kg piston is attached to a spring and exposed to an atmospheric pressure of 1 bar. Initially, the bottom of the piston is at $x = 0$, and the spring exerts a negligible force on the piston. The valve is opened and air enters the cylinder from the supply line, causing the volume of the air within the cylinder to increase by $3.9 \times 10^{-4} \text{ m}^3$. The force exerted by the spring as the air expands within the cylinder varies linearly with x according to $F_{\text{spring}} = kx$, where $k = 10,000 \text{ N/m}$. The piston face area is $7.8 \times 10^{-3} \text{ m}^2$. Ignoring friction between the piston and the cylinder wall, determine the pressure of the air within the cylinder, when the piston is in its initial position. Repeat when the piston is in its final position. The atmospheric pressure is 100 kPa and the local acceleration of gravity is 9.81 m/s^2 .

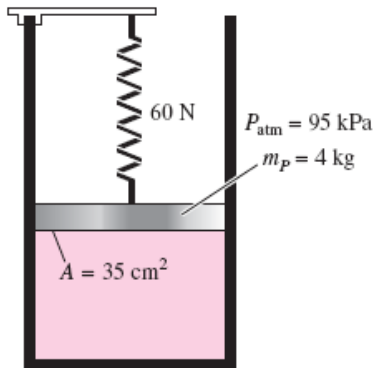


Figure P1.9

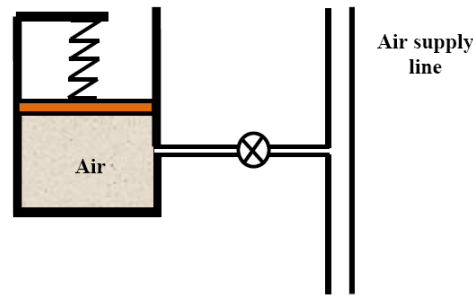


Figure P1.10/11

12. A new scale N of temperature is devised in such a way that the freezing point of ice is 20^0N and boiling point is 200^0N . What will be the temperature reading on 0C , 0F and K scales when the temperature on the new scale is 160^0N ?

TUTORIAL NO: 2

ENERGY TRANSFER

1. A gas undergoes compression from an initial state of $V_1 = 0.1 \text{ m}^3$, $P_1 = 200 \text{ kPa}$ to a final state of $V_2 = 0.04 \text{ m}^3$, $P_2 = 500 \text{ kPa}$. If the pressure varies linearly with volume during the process, determine the work transfer.
2. A gas undergoes a ploytropic process from an initial state of 500 kPa and 0.02 m^3 to a final state of 100 kPa and 0.05 m^3 . Determine the work transfer.
3. In a non flow process, a gas expands from volume 0.1 m^3 to a volume of 0.2 m^3 according to the law

$$P = \frac{2}{V} + 1.5$$

where P is the pressure in bar, and V is the volume in m^3 . Determine (i) the pressure at the end of the expansion and (ii) the work done by the gas in the expansion process, in kJ.

4. Consider a two-part process with an expansion from 0.1 m^3 to 0.2 m^3 at a constant pressure of 150 kPa followed by an expansion from 0.2 m^3 to 0.4 m^3 with a linearly rising pressure from 150 kPa ending at 300 kPa. Show the process in a P - V diagram and find the boundary work.
5. Air undergoes three process in series to form a cycle:
 - Process 1-2:** constant volume cooling, $V_1 = V_2 = 2 \text{ m}^3$
 - Process 2-3:** constant pressure heating, $P = 100 \text{ kPa}$, $V_3 = 10 \text{ m}^3$
 - Process 3-1:** isothermal compression

Sketch the process on P - V diagram and determine the total work transfer.

6. Argon (1 kg) contained in a piston cylinder device shown in **Figure P2.6** is initially at a pressure of 500 kPa and a temperature of 70^0C . Heat is added until the final temperature is 600^0C and a pressure of 1 MPa is required to lift the piston from the stops. Sketch the process on P - V and T - V diagrams and determine the total work transfer. [Take $R = 208 \text{ J/kg K}$]
7. Air (0.5 kg) in the piston cylinder device shown in **Figure P2.7** has an initial pressure and temperature of 1 MPa and 500^0C respectively. The system is cooled until the temperature reaches 50^0C . It takes a pressure of 0.5 MPa to support the piston. Sketch the process on P - V and T - V diagrams and determine the total work transfer. [Take $R = 287 \text{ J/kg K}$]
8. Oxygen (3.6 kg) contained in a piston cylinder device shown in **Figure P2.8** is initially at a pressure of 200 kPa and a temperature of 50^0C . Heat is added until the piston just reaches the upper stops where the total volume is 3 m^3 . It

requires a pressure of 500 kPa to lift the piston. Sketch the process on P-V and T-V diagrams and determine the total work transfer. [Take $R = 260 \text{ J/kg K}$]

9. A piston cylinder device shown in **Figure P2.9** contains 1 kg of Nitrogen initially at a pressure of 250 kPa and a temperature of 500°C . Heat is lost from the system till its temperature reaches 25°C . Sketch the process on P-V and T-V diagrams and determine the final pressure and the total work transfer. [Take $R = 297 \text{ J/kg K}$]

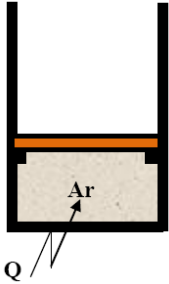


Figure P2.6



Figure P2.7

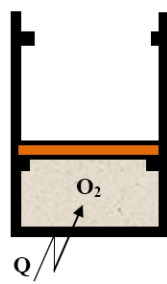


Figure P2.8

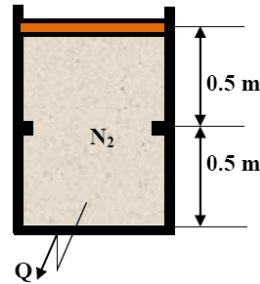


Figure P2.9

10. A piston cylinder arrangement loaded with a linear spring ($k = 2 \text{ kN/m}$) as shown in **Figure P2.10** contains air. Spring is initially unstretched and undergoes a compression of 40 mm during a process. If the mass of the piston is 80 kg and piston diameter is 0.1 m, determine the total work transfer. [Take $P_{atm} = 100 \text{ kPa}$ and $g = 9.81 \text{ m/s}^2$]
11. Air (0.01 kg) is contained in a piston cylinder device restrained by a linear spring ($k = 500 \text{ kN/m}$) as shown in **Figure P2.11**. Spring initially touches the piston but exerts no force on it. Heat is added to the system until the piston is displaced upward by 80 mm. determine
 (a) the temperature at which piston leaves the stops
 (b) work done by the air [Take $R = 287 \text{ J/kg K}$, $P_{atm} = 100 \text{ kPa}$ and $g = 9.81 \text{ m/s}^2$]
12. A piston cylinder arrangement shown in **Figure P2.12** is restrained by two linear springs as shown. The system contains air initially at a pressure of 150 kPa and a volume of 0.002 m^3 . Heat is added to the system until its volume doubles; determine the total work transfer. Also sketch the process on P-V diagram. Both springs have spring constant of 100 kN/m.

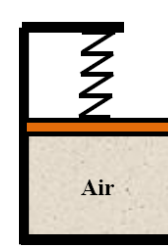


Figure P2.10

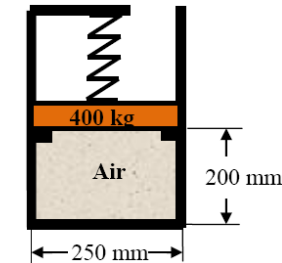


Figure P2.11

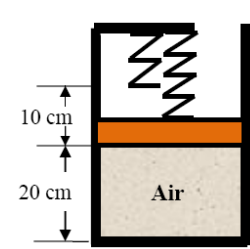


Figure P2.12

TUTORIAL NO: 3

PROPERTIES OF COMMON SUBSTANCES

1. Fill in the blanks in the following table with the corresponding properties of water or by the symbol $\times$, when it is no relevant or meaningless or by the symbol $-$, when it is indeterminate.

State	P, kPa	T, °C	x, %	v, m ³ /kg	h, kJ/kg	Degree of Superheat
1	200	200				
2	250		80			
3	300			0.5		
4		150	60			
5		200		0.1		
6		250		0.5951		
7	1000				2000	
8		350			2563.5	
9	600			0.001101		
10	2000			0.1757		
11	5000			0.001206		
12		500			3445.4	
13	8000	295.04				

2. A two phase mixture of H₂O has a temperature of 200°C and occupies a volume of 0.05 m³. The mass of saturated liquid is 1 kg and saturated vapor is 3 kg. Determine the pressure and specific volume of the mixture.
3. Determine the temperature and quality (if needed) for water at a pressure of 200 kPa and having a specific volume of
 (a) 0.8 m³/kg
 (b) 1.25 m³/kg.
4. Water is contained in a rigid vessel of 5 m³ at a quality of 0.8 and a pressure of 2 MPa. If it is cooled to a pressure of 400 kPa, determine the mass of saturated liquid and saturated vapor at the final state.
5. A piston cylinder device shown in **Figure P3.5** contains 0.2 kg of a mixture of saturated liquid water and saturated water vapor at a temperature of 50°C and a volume of 0.03 m³. The mass of the piston resting on the stops is 50 kg and the cross sectional area of the piston is 12.2625 cm². The atmospheric pressure is 100 kPa. Heat is transferred until it becomes saturated vapor. Sketch the process on P-v and T-v diagrams and determine:
 (a) the temperature at which the piston just leaves the stops,
 (b) the final pressure, and
 (c) the total work transfer. [Take $g = 9.81 \text{ ms}^{-2}$]

6. A piston cylinder device shown in **Figure P3.6** contains water initially at a pressure of 125 kPa with a quality of 50 %. Heat is added to the system until it reaches to a final temperature of 800°C. It takes a pressure of 600 kPa to lift the piston from the stops. Sketch the process on P-v and T-v diagrams and determine:
 (a) the mass of H₂O in the system, and
 (b) the total work transfer.
7. The frictionless piston shown in **Figure P3.7** has a mass of 20 kg and a cross sectional area of 78.48 cm². Heat is added until the temperature reaches 400°C. If the quality of the H₂O at the initial state is 0.2, determine:
 (a) the initial pressure,
 (b) the mass of H₂O,
 (c) the quality of the system when the piston hits the stops,
 (d) the final pressure, and
 (e) the total work transfer. [Take $P_{\text{atm}} = 100 \text{ kPa}$, $g = 9.81 \text{ m/s}^2$]
8. Water (2 kg) is contained within a piston-cylinder arrangement as shown in **Figure P3.8**. The initial temperature is 105°C, and there are equal masses of liquid and vapor initially. The system is heated to a position where the piston is locked, and then the system is cooled to the saturated vapor state at $T = 50^\circ\text{C}$. Draw the process on a P-v diagram, and evaluate the work done during the process.



Figure P3.5

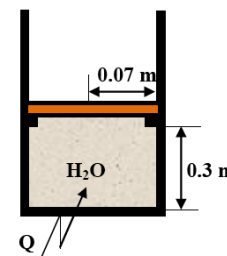


Figure P3.6

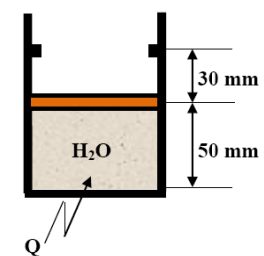


Figure P3.7

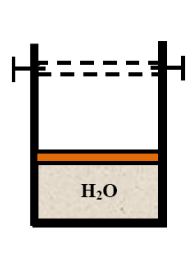


Figure P3.8

9. A piston cylinder arrangement shown in **Figure P3.9** contains 1 kg of water initially at a pressure of 1 MPa and a temperature of 500°C. The water is cooled until it is completely converted into the saturated liquid. It requires a pressure of 400 kPa to support the piston. Sketch the process on P-v and T-v diagrams and determine the total work transfer.
10. A piston cylinder arrangement shown in **Figure P3.10** contains 2 kg of water initially at a pressure of 200 kPa and a temperature of 50°C. Heat is added until the piston reaches the upper stops where the total volume is 1.5 m³. It takes a pressure of 600 kPa to lift the piston. Sketch the process on P-v and T-v diagrams and determine the final temperature and the work transfer.

11. A piston cylinder device with a linear spring initially contains water at a pressure of 4 MPa and 500°C with the initial volume being 0.1 m³, as shown in **Figure P3.11**. If the piston is at the bottom, the system pressure is 300 kPa. The system now cools until the pressure reaches 1000 kPa. Sketch the process on P-v diagram and determine

- the mass of H₂O
- the final temperature and volume, and
- the total work transfer.

12. A piston cylinder arrangement shown in **Figure P3.12** contains water initially at $P_1 = 100$ kPa, $x_1 = 0.8$ and $V_1 = 0.01$ m³. When the system is heated, it encounters a linear spring ($k = 100$ kN/m). At this state volume is 0.015 m³. The heating continues till its pressure is 200 kPa. If the diameter of the piston is 0.15 m, determine

- the final temperature, and
- the total work transfer.

Also sketch the process on P-v diagram.

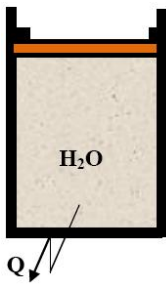


Figure P3.9

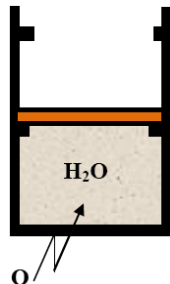


Figure P3.10

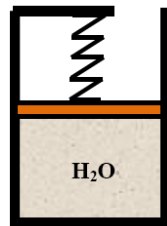


Figure P3.11

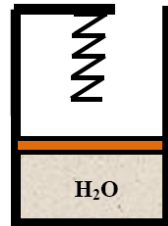


Figure P3.12

TUTORIAL NO: 4

FIRST LAW OF THERMODYNAMICS

- A piston cylinder device contains 0.2 kg of a gas initially at $P_1 = 1000$ kPa and $V_1 = 0.02$ m³. It undergoes polytropic expansion to a final pressure of 200 kPa during which the relation between pressure and volume is $PV^3 = \text{constant}$. If the specific internal energy of the gas decreases by 160 kJ/kg during the process, determine the heat transfer for the process.
- A rigid vessel with a volume of 0.1 m³ contains water initially at 500 kPa with a quality of 60 %. A heater is turned on heating the water at a rate of 2 kW. Determine the time required to vaporize all the liquid.
- A piston cylinder device shown in **Figure P4.3** restrained by a linear spring contains 2 kg of air initially at 150 kPa and 27°C. It is now heated until its volume doubles at which time temperature reaches 527°C. Sketch the process on $P-v$ and determine the total work and heat transfer in the process. [Take $R = 287$ J/kgK and $c_v = 718$ J/kgK].
- Nitrogen (5 kg) is contained in a piston cylinder device shown in **Figure P4.4** initially at a pressure of 800 kPa and a temperature of 127°C. There is a heat transfer to the system until the temperature reaches to 527°C. It takes a pressure of 1500 kPa to lift the piston. Sketch the process on $P-V$ and $T-V$ diagrams and determine the total work and heat transfer in the process. [Take $R = 297$ J/kgK and $c_v = 743$ J/kgK].
- Air (0.4 kg) is contained in a piston cylinder device shown in **Figure P4.5** initially at a pressure of 1500 kPa and 800 K. The cylinder has stops such that the minimum volume of the system is 0.04 m³. The air in the cylinder is cooled to 300 K. Sketch the process on $P-V$ and $T-V$ diagrams and determine
 - the final volume and pressure of the air, and
 - the total work and heat transfer in the process. [Take $R = 287$ J/kgK and $c_v = 718$ J/kgK].
- Air (1 kg) is contained in piston/cylinder assembly as shown in **Figure P4.6**. Initially, the piston rests on the stops and is in contact with the spring, which is in its unstretched position. The spring constant is 100 kN/m. The piston weighs 30 kN and atmospheric pressure is 101 kPa. The air is initially at 300 K and 200 kPa. Heat transfer occurs until the air temperature reaches the surrounding temperature of 700 K.
 - Find the final pressure and volume.
 - Find the process work.
 - Find the heat transfer.

(d) Draw the P-V diagram of the process. [Take $R = 287 \text{ J/kgK}$ and $c_v = 718 \text{ J/kgK}$].

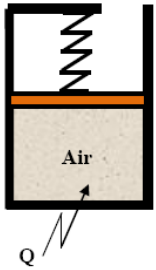


Figure P4.3

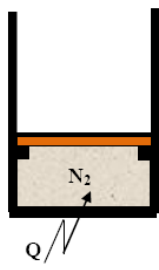


Figure P4.4



Figure P4.5

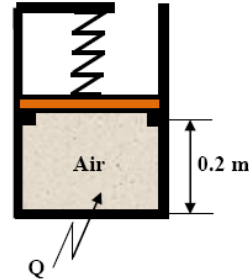


Figure P4.6

7. Water (1.5 kg) is contained in a piston cylinder device shown in **Figure P4.7** initially at a pressure of 400 kPa with a quality of 50% . There is a heat transfer to the system until it reaches a final temperature of 500°C . It takes a pressure of 800 kPa to lift the piston. Sketch the process on $P-v$ and $T-v$ diagrams and determine the total work and heat transfer.
8. A piston cylinder device shown in **Figure P4.8** contains 4 kg of water initially at saturated liquid state at 5 MPa . There is a heat transfer to the system until it hits the stops at which time its volume is 0.08 m^3 . There is further heat transfer to the device until water is completely vaporized. Sketch the process on $P-v$ and $T-v$ diagrams and determine the total work and heat transfer.
9. A piston cylinder device shown in **Figure P4.9** contains water initially at $P_1 = 1 \text{ MPa}$ and $T_1 = 500^\circ\text{C}$. A pressure of 400 kPa is required to support the piston. There is a heat transfer from the device until its temperature drops to 30°C . Sketch the process on $P-v$ and $T-v$ diagrams and determine the total work and heat transfer.

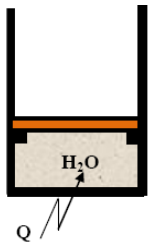


Figure P4.7

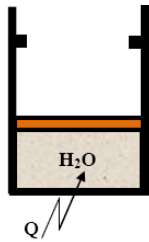


Figure P4.8

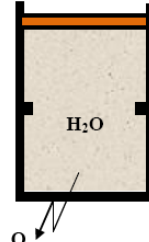


Figure P4.9

10. A closed system undergoes a process A from state 1 to state 2 as shown in **Figure P4.10**; which requires a heat input of $Q_A = 65 \text{ kJ}$. The system returns adiabatically from state 2 to state 1 through process B . Determine the work transfer for process B .

11. Air enters a compressor operating at steady state at 100 kPa , 300 K and leaves at 1000 kPa , 400 K , with a volumetric flow rate of $1.5 \text{ m}^3/\text{min}$. The work consumed by the compressor is 250 kJ per kg of air. Neglecting the effects of potential and kinetic energy, determine the heat transfer rate, in kW . [Take $R = 287 \text{ J/kgK}$ and $c_p = 1005 \text{ J/kgK}$]

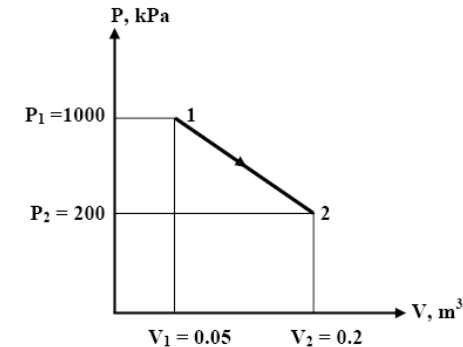


Figure P4.10

12. A steam turbine develops 60 MW of power output. Mass flow rate of steam is found to be 80 kg/s . Properties of steam at inlet and exit of the turbine are as follows:

Property	Inlet	Exit
Pressure	8 MPa	0.4 MPa
Temperature	500°C	
Quality		80%
Velocity	50 m/s	150 m/s
Elevation above the reference level	10 m	5 m

- (a) Determine the rate at which heat is lost from the turbine surface.
- (b) Determine the inlet and outlet areas.

13. Air enters a nozzle steadily at 300 kPa , 127°C and with a velocity of 40 m/s and leaves at 100 kPa and with a velocity of 300 m/s . The heat loss from the nozzle surface is 20 kJ/kg of the air. The inlet area of the nozzle is 100 cm^2 . Determine:

- (a) the exit temperature of the air, and
- (b) the exit area of the nozzle.

[Take $R = 287 \text{ J/kgK}$ and $c_p = 1005 \text{ J/kgK}$]

14. In a water heat operating under steady state condition, water at 50°C flowing with a mass flow rate of 5 kg/s is mixed with the saturated vapor at 120°C . The mixture the heater as a saturated liquid water at 100°C . Determine the rate at which saturated water vapor must be supplied to the heater.
15. Steam enters into a well insulated throttling valve at 10 MPa , 600°C and exits at 5 MPa . Determine the final temperature of the steam.

TUTORIAL NO: 5

SECOND LAW OF THERMODYNAMICS

- An inventor makes the following claims. Determine whether the claims are valid or not and explain why or why not.
 - A petrol engine operating between temperatures 2000°C and 500°C will produce 1.2 kW of power output consuming 0.15 kg/h of petrol having a calorific value of 42500 kJ/kg .
 - A heat pump supplies heat to a room maintained at 22°C at a rate of 50000 kJ/h . The inventor claims a work input of 5000 kJ/h is sufficient when the surroundings is at -2°C .
 - A refrigerator maintains -5°C in the refrigerator which is kept in a room where the temperature is 30°C and has a COP of 8.
- A power cycle operating between two reservoirs receives Q_H from a high temperature source at $T_H = 1000\text{ K}$ and rejects energy Q_L to a low temperature sink at $T_L = 300\text{ K}$. For each of the following cases, determine whether the cycle operates reversibly, irreversibly or is impossible.
 - $Q_H = 800\text{ kJ}$, $W = 600\text{ kJ}$
 - $Q_H = 800\text{ kJ}$, $Q_L = 240\text{ kJ}$
 - $W = 960\text{ kJ}$, $Q_L = 640\text{ kJ}$
 - $\eta = 50\%$
- A car engine consumes fuel at a rate of 30 L/h and delivers 80 kW of power output. If the calorific value of the fuel is 42000 kJ/kg and a density of 0.8 g/cm^3 , determine the efficiency of the engine.
- A Carnot engine operates between two reservoirs at temperatures T_L and T_H . The work output of the engine is 0.6 times the heat rejected. The difference in temperatures between the source and the sink is 200°C . Calculate the thermal efficiency, source temperature and the sink temperature.
- A refrigerator having a COP of 4 maintains the freezer compartment at -3°C by removing heat at a rate of 10800 kJ/h and rejects heat to the surroundings at 27°C .
 - Determine the power input to the refrigerator and compare it with minimum theoretical power input.
 - If the electricity costs Rs $10/\text{kWh}$, determine the actual and minimum theoretical cost per day for effective operation of 12 h/day .
- A heat pump heats a house in the winter and then reverses to cool it in the summer. The room temperature should be 22°C in the winter and 26°C in the summer. Heat transfer through the walls and ceilings is estimated to be 3000 kJ/h per degree temperature difference between the inside and outside.
 - Determine the power required to run it in the winter when the outside temperature decrease to 0°C .

- If the unit is run by the same power as calculated in (a), throughout the year, determine the maximum outside summer temperature for which the house can be maintained at 26°C .
- A piston cylinder device shown in **Figure P5.7** contains 1.5 kg of water initially at 100 kPa with 10 % of quality. The mass of the piston is such that a pressure of 500 kPa is required to lift the piston. Heat is added to the system from a source at 500°C until its temperature reaches 400°C . Determine the total entropy generation during the process.

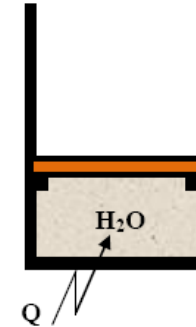


Figure P5.7

- 2 kg water at 100°C is mixed with 4 kg of water at 20°C in an isolated system. Calculate the net change in entropy due to the mixing process. [Take specific heat of water $c = 4.18\text{ kJ/K}$]
- Two blocks A and B, which are initially at 95°C and 540°C , respectively are brought together into contact and isolated from the surroundings. They are allowed to reach a final state of thermal equilibrium. Determine the entropy change of each block and of the isolated system. Block A is aluminium ($c_p = 0.9\text{ kJ/kg K}$) with $m_A = 0.45\text{ kg}$, and block B is copper ($c_p = 0.385\text{ kJ/kg K}$) with $m_B = 0.9\text{ kg}$.
- 1 kg of air enclosed in an isolated box with volume V_1 , pressure P_1 and temperature T_1 is allowed to expand freely until its volume increases to $V_2 = 2V_1$. Determine the change in entropy. [Take $R = 287\text{ J/kgK}$]
- Steam enters the nozzle at 1 MPa , 300°C , with a velocity of 30 m/s . The pressure of the steam at the nozzle exit is 0.3 MPa . Determine the exit velocity of the steam from the nozzle, assuming a reversible, adiabatic, steady state, steady flow process.
- Air at 200 kPa and 1000 K with very low velocity enters into a nozzle and exits at a pressure of 100 kPa . Assuming the process to be isentropic, determine the exit velocity.
- Steam enters into a well insulated throttling valve at 10 MPa and 600°C and exits at 5 MPa . Determine the change in entropy per unit mass of the steam.

14. Steam enters into a turbine at a rate of 2 kg/s with $P_1 = 2 \text{ MPa}$, $T_1 = 750^\circ\text{C}$ and exits at $P_2 = 10 \text{ kPa}$.
- If the turbine is isentropic, what is the power output of the turbine?
 - If the isentropic efficiency of the turbine is 80 %, what is the power output?
 - What is the outlet enthalpy of the steam from the real turbine?
15. Air enters a gas turbine at 1 MPa and 1500 K and exits at 100 kPa . If its isentropic efficiency is 80 %, determine the turbine exit temperature.

TUTORIAL NO: 6

THERMODYNAMIC CYCLES

- Air at the compressor inlet of an ideal gas turbine cycle is at 100 kPa and 20°C . The heat added to the cycle per kg of air is 800 kJ/kg . The maximum temperature during the cycle is limited to 1400 K . Determine
 - the pressure ratio,
 - the net work output per kg of air, and
 - the cycle efficiency.
- In an ideal Brayton cycle, air enters the compressor at 100 kPa and 300 K and the turbine at 1000 kPa and 1200 K . Heat is transferred to the air at a rate of 30 MW . Determine the efficiency and the power output of the plant.
- The compressor and turbine of an ideal gas turbine each have isentropic efficiencies of 80 %. The pressure ratio is 10. The minimum and maximum temperatures are 300 K and 1200 K respectively. Determine:
 - the net work per kg of air,
 - the thermal efficiency of the cycle, and
 - Compare both of these for a cycle with ideal compressor and turbine.
- The compression ratio of an ideal Otto cycle is 8.5. At the beginning of the compression stroke, air is at 100 kPa and 27°C . The pressure is doubled during the constant volume heat addition process. Determine:
 - the heat added per kg of air,
 - the net work output per kg of air,
 - the thermal efficiency, and
 - the mean effective pressure..
- The following data are obtained for a four stroke petrol engine:

Cylinder bore	= 14 cm
Stroke length	= 15 cm
Clearance volume	= 231 cm^3

 Determine:
 - the ratio of clearance volume and swept volume,
 - the compression ratio, and
 - the thermal efficiency.
- An engine with bore of 8 cm and stroke of 12 cm has a compression ratio of 6. To increase the compression ratio 1.5 mm is machined off the cylinder head face. Determine the new compression ratio.
- The properties of air at the beginning of compression stroke in an air standard Diesel cycle are 100 kPa and 300 K . The air at the beginning of the expansion stroke is at 6500 kPa and 2000 K . Determine:
 - the compression ratio,

- (b) the thermal efficiency, and
 - (c) the mean effective pressure.
8. An air standard diesel cycle has a compression ratio of 22 and expansion ratio of 11. Determine its cut off ratio and the efficiency.
 9. Air at the beginning of compression stroke in an ideal Diesel cycle is at 100 kPa and 295 K and the compression ratio is 20. Determine the maximum temperature during the cycle to have an efficiency of 65 %.
 10. A Rankine cycle has a boiler working at a pressure of 2 MPa . The maximum and minimum temperatures during the cycle are 400°C and 50°C respectively. Determine the efficiency of the cycle and compare it with that of the Carnot cycle operating between the same temperature limits.
 11. Saturated vapor enters into a turbine of an ideal Rankine cycle at 10 MPa and saturated liquid exits the condenser at 10 kPa . The power output of the cycle is 120 MW . Determine:
 - (a) the thermal efficiency of the cycle,
 - (b) the back work ratio,
 - (c) the mass flow rate of steam,
 - (d) the rate at which heat is supplied to the boiler,
 - (e) the rate at which heat is rejected from the condenser, and
 - (f) the mass flow rate of condenser cooling water, if the cooling water enters at 20°C and exits at 35°C . [Take specific heat of water as 4.18 kJ/kgK].
 12. Superheated steam at 8 MPa , 500°C enters into turbine of a steam power plant working on a Rankine cycle. The steam leaves the condenser as saturated liquid at 8 kPa . The turbine and pump have isentropic efficiencies of 90 % and 80 % respectively. For the cycle, determine:
 - (a) the net work per kg of steam,
 - (b) the heat supplied into the boiler per kg of steam, and
 - (c) the thermal efficiency.

TUTORIAL NO: 7

HEAT TRANSFER

1. Determine the rate of heat loss from a brick wall ($k = 0.7 \text{ W/mK}$) of length 5 m , height 4 m and 0.25 m thick. The temperature of the inner surface is 120°C and that of outer surface is 30°C . Also calculate the distance from the inner surface at which temperature is 90°C .
2. The roof of an electrically heated home is 10 m long, 8 m wide, and 0.25 m thick, and is made of a flat layer of concrete whose thermal conductivity is $k = 0.8 \text{ W/mK}$. The temperatures of the inner and the outer surfaces of the roof one night are measured to be 18°C and 5°C , respectively, for a period of 12 hours . Determine:
 - (a) the rate of heat loss through the roof that night and
 - (b) the cost of that heat loss to the home owner if the cost of electricity is $\text{Rs } 10/\text{kWh}$.
3. An electric current is passed through a wire 2 mm in diameter and 8 cm long. The wire is submerged in the liquid water. During the boiling of water temperature of water is 100°C and convection heat transfer coefficient is $4500 \text{ W/m}^2\text{K}$. Determine the power supplied to the wire to maintain the wire surface temperature at 120°C .
4. The inner surfaces of the walls, floors, and the ceiling of the house are observed to be at an average temperature of 5°C in winter and 26°C in summer. Determine the rate of radiation heat transfer between a person inside the house and the surrounding surfaces if the exposed surface area, the average outer surface temperature and the surface emissivity of the person are 1.4 m^2 and 30°C and 0.95, respectively.
5. A flat plate solar collector is insulated at the back surface and exposed to solar radiation at the front surface. The front surface absorbs solar radiation at a rate of 800 W/m^2 and losses heat by convection to the ambient air at 25°C . If the heat transfer coefficient between the plate and the air is $20 \text{ W/m}^2\text{K}$, determine the surface temperature of the plate.
6. The inner surface of a 2 cm thick $50 \text{ cm} \times 50 \text{ cm}$ plate ($k = 10 \text{ W/mK}$) is at 400°C . The outer surface dissipates heat by combined convection and radiation to the ambient air at 27°C . If the plate surface has an emissivity 0.85 and the convection heat transfer coefficient between the outer plate surface and the ambient air is $20 \text{ W/m}^2\text{K}$, determine the outer surface temperature of the plate.
7. The walls of a furnace $4 \text{ m} \times 3 \text{ m}$ are constructed from an inner fire brick ($k = 0.4 \text{ W/mK}$) wall 30 cm thick, a layer of ceramic blanket insulation ($k = 0.2 \text{ W/mK}$) 10 cm thick and steel protective layer ($k = 50 \text{ W/mK}$)

4 mm thick. The inside temperature of the fire brick layer was measured as 500°C and the temperature of the outside of the insulation as 50°C . Determine

- the rate of heat loss through the wall,
- the temperature at the interface between fire brick layer and insulation layer, and
- the temperature at the outside surface of the steel layer.

8. Find the heat transfer through the composite wall as shown in **Figure P7.8**. Assume one dimensional flow. The thermal conductivities of wall materials are $k_A = 150 \text{ W/mK}$, $k_B = 30 \text{ W/mK}$, $k_C = 65 \text{ W/mK}$ and $k_D = 50 \text{ W/mK}$. All dimensions are in cm.

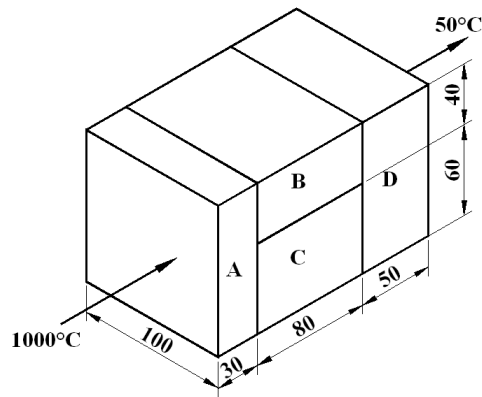


Figure P7.8

- An exterior wall of a house consists of 10 cm of common brick ($k = 0.8 \text{ W/mK}$) followed by a 4 cm layer of gypsum plaster ($k = 0.5 \text{ W/mK}$). What thickness of rock wool insulation ($k = 0.065 \text{ W/mK}$) should be added to reduce the heat transfer through the wall by 50 %?
- A cast iron pipe ($k = 25 \text{ W/mK}$) with inner and outer diameters of 60 mm and 70 mm respectively is covered by an insulator ($k = 0.05 \text{ W/mK}$). Under steady state condition, temperature between the pipe and insulator interface is found to be 250°C . The allowable heat loss from the unit length of the pipe is 500 W and outer surface temperature of the insulator should not exceed 50°C . Determine:
 - minimum thickness of the insulation required, and
 - temperature at the inner surface of the pipe.
- A gas turbine blade is modeled as a flat plate. The thermal conductivity of the blade material is 15 W/mK and its thickness is 1.5 mm. The upper surface of the blade is exposed to hot gases at 1000°C and the lower surface is cooled by air bled of the compressor. The heat transfer coefficients at the upper and lower surfaces of the blade are $2500 \text{ W/m}^2\text{K}$ and $1500 \text{ W/m}^2\text{K}$

respectively. Under steady state conditions, the temperature, at the upper surface of the blade is measured as 850°C , determine the temperature of the coolant air.

- The inside surface of an insulating layer is at 300°C and the outside surface is dissipating heat by convection into air at 25°C . The insulating layer has a thickness 5 cm of and thermal conductivity of 0.8 W/mK . What is the minimum heat transfer coefficient at the outside surface if the outside surface temperature should not exceed 100°C ?

ANSWERS

TUTORIAL NO: 1

1. $0.96154 \text{ m}^3/\text{kg}$
2. $10 \text{ m}^3/\text{kg}$
3. $8 \text{ m}^3/\text{kg}, 2.667 \text{ m}^3$
4. $2 \text{ m}^3/\text{kg}, 5 \text{ m}^3/\text{kg}$
5. $127.67 \text{ kPa}, 267.67 \text{ kPa}$
6. $220 \text{ kPa}, 400 \text{ kPa}, 300 \text{ kPa}$
7. 178.48 kPa
8. 121.3 kPa
9. 123.4 kPa
10. 469.357 kPa
11. $112.6 \text{ kPa}, 176.7 \text{ kPa}$
12. $77.778 ^\circ\text{C}, 172 ^\circ\text{F}, 350.778 \text{ K}$

TUTORIAL NO: 2

1. -21 kJ
2. 6.609 kJ
3. $250 \text{ kPa}, 288.63 \text{ kJ}$
4. 60 kJ
5. -809.438 kJ
6. 38.896 kJ
7. -9.112 kJ
8. 744.18 kJ
9. $192.76 \text{ kPa}, -114.79 \text{ kJ}$
10. 81 J
11. $342.52 ^\circ\text{C}, 1.31 \text{ kJ}$
12. 2.8 kJ

TUTORIAL NO: 3

1.	Stat e	P, kPa	T, $^\circ\text{C}$	x, %	v, m^3/kg	h, kJ/kg	Degree of Superheat, $^\circ\text{C}$
	1	200	200	x	1.0803	2870.0	79.76
	2	250	127.44	80	0.575227	2280.53	x
	3	300	133.56	82.495	0.5	2346.544	x
	4	475.72	150	60	0.23617	1900.78	x
	5	1553.6	200	78.385	0.1	2373.135	x
	6	400	250	x	0.5951	2963.6	106.36
	7	1000	179.92	61.402	0.11982	2000	x
	8	16520	350	100	0.008812	2563.5	0
	9	600	158.86	0	0.001101	670.71	x
	10	2000	500	x	0.1757	3467.7	287.58
	11	5000	230	x	0.001206	990.43	x
	12	4000	500	x	0.08642	3445.4	249.61
	13	8000	295.04	—	—	—	—

2. $1553.6 \text{ kPa}, 0.095731 \text{ m}^3/\text{kg}$
3. $120.24 ^\circ\text{C}, 0.90296; 271.806 ^\circ\text{C}$
4. $10.6896 \text{ kg}, 51.8847 \text{ kg}$
5. $151.87 ^\circ\text{C}, 500 \text{ kPa}, 22.49 \text{ kJ}$
6. $0.006714 \text{ kg}, 0.5497 \text{ kJ}$
7. $125 \text{ kPa}, 0.0014 \text{ kg}, 0.321, 711.71 \text{ kPa}, 0.0294 \text{ kg}$
8. 2736.24 kJ
9. -141.2 kJ
10. $703.618 ^\circ\text{C}, 898.785 \text{ kJ}$
11. $1.1571 \text{ kg}; 179.92 ^\circ\text{C}, 0.01892 \text{ m}^3; -202.703 \text{ kJ}$

12. $627.165 ^\circ\text{C}, 0.54684 \text{ kJ}$

TUTORIAL NO: 4

1. -25.42 kJ
2. $2 \text{ min } 54.4 \text{ s}$
3. $200.9 \text{ kJ}, 918.9 \text{ kJ}$
4. $74.25 \text{ kJ}, 1510.25 \text{ kJ}$
5. $0.04 \text{ m}^3, 861 \text{ kPa}; -31.84 \text{ kJ}, -175.44 \text{ kJ}$
6. $294.63 \text{ kPa}, 0.0682 \text{ m}^3, 6.72 \text{ kJ}, 35.44 \text{ kJ}$
7. $87.7152 \text{ kJ}, 1531.3152 \text{ kJ}$
8. $374.28 \text{ kJ}, 6002.98 \text{ kJ}$
9. $-176.55 \text{ kJ}, -3175.38 \text{ kJ}$
10. -25 kJ
11. $4.998 \text{ kg/s}, 760.191 \text{ m/s}, 8.2459 \text{ cm}^2$
12. $-26.14 \text{ MW}, 0.066784 \text{ m}^2, 0.197442 \text{ m}^2$
13. $151.08 ^\circ\text{C}, 42.41 \text{ cm}^2$
14. 0.4585 kg/s
15. $582.112 ^\circ\text{C}$

TUTORIAL NO: 5

1. Invalid, Valid, Invalid
2. Impossible, reversible, irreversible, irreversible
3. 28.57%
4. $533.3 \text{ K}, 333.3 \text{ K}$
5. $0.75 \text{ kW}, 0.3297 \text{ kW}; \text{Rs } 90, \text{Rs } 39.56$
6. $1.367 \text{ kW}, 48.148 ^\circ\text{C}$
7. 3.8485 kJ/K
8. 1.1913 kJ/K
9. $0.179 \text{ kJ/K}, -0.121 \text{ kJ/K}, 0.0583 \text{ kJ/K}$
10. 198.93 J/K
11. 737 m/s
12. 600.9375 m/s
13. 0.3075 kJ/kg K
14. $2214.266 \text{ kW}, 1771.413 \text{ kW}, 2852.56 \text{ kJ/kg}$
15. 921.537 K

TUTORIAL NO: 6

1. $12.58, 411.91 \text{ kJ/kg}, 51.49 \%$
2. $48.205 \%, 14.46 \text{ MW}$
3. $230.599 \text{ kJ/kg}, 26.96 \%; 446.089 \text{ kJ/kg}, 48.205 \%$
4. $507.01 \text{ kJ/kg}, 291.61 \text{ kJ/kg}, 57.52 \%, 383.84 \text{ kPa}$
5. $0.1, 11, 61.68 \%$
6. 6.41
7. $19.72, 64.38 \%, 800.47 \text{ kPa}$
8. $2, 65.99 \%$
9. 1886 K
10. $31.635 \%, 52.01 \%$
11. $29.618 \%, 1.063 \%, 127.8 \text{ kg/s}, 322.38 \text{ MW}, 202.38 \text{ MW}, 3227.82 \text{ kg/s}$
12. $1155.54 \text{ kJ/kg}, 3214.58 \text{ kJ/kg}, 35.95 \%$

TUTORIAL NO: 7

1. $5.04 \text{ kW}, 8.33 \text{ cm}$
2. $3.328 \text{ kW}, \text{Rs } 332.8$
3. 45.2389 W
4. $185.215 \text{ W}, 32.906 \text{ W}$
5. $65 ^\circ\text{C}$
6. 643.511 K
7. $3600 \text{ W}, 275 ^\circ\text{C}, 49.98 ^\circ\text{C}$
8. 20.53 kW
9. 1.3325 cm
10. $4.687 \text{ mm}, 250.491 ^\circ\text{C}$
11. $562.5 ^\circ\text{C}$
12. $42.667 \text{ W/m}^2\text{K}$

APPENDIX 2

Table A 2.1: Properties of SATURATED WATER – Pressure Table

P kPa	T °C	v_l m ³ /kg	v_{lg} m ³ /kg	v_g m ³ /kg	u_l kJ/kg	u_{lg} kJ/kg	u_g kJ/kg	h_l kJ/kg	h_{lg} kJ/kg	h_g kJ/kg	s_l kJ/kg.K	s_{lg} kJ/kg.K	s_g kJ/kg.K
1.0	6.9696	0.001	129.19	129.19	29.287	2354.8	2384.1	29.288	2484.0	2513.3	0.1059	8.8678	8.9737
1.5	13.021	0.001001	87.970	87.971	54.634	2337.9	2392.5	54.635	2469.8	2524.4	0.1954	8.6304	8.8258
2.0	17.497	0.001001	66.997	66.998	73.364	2325.2	2398.6	73.366	2459.2	2532.6	0.2603	8.4613	8.7216
2.5	21.080	0.001002	54.248	54.249	88.353	2315.1	2403.5	88.356	2450.7	2539.1	0.3116	8.3295	8.6411
3.0	24.083	0.001003	45.660	45.661	100.92	2306.7	2407.6	100.92	2443.7	2544.6	0.3541	8.2214	8.5755
3.5	26.677	0.001003	39.473	39.474	111.77	2299.4	2411.2	111.77	2437.5	2549.3	0.3904	8.1299	8.5203
4.0	28.966	0.001004	34.797	34.798	121.34	2293.0	2414.3	121.35	2432.2	2553.5	0.4222	8.0503	8.4725
4.5	31.018	0.001005	31.136	31.137	129.93	2287.2	2417.1	129.93	2427.3	2557.2	0.4506	7.9799	8.4305
5.0	32.881	0.001005	28.190	28.191	137.72	2281.9	2419.6	137.72	2422.8	2560.5	0.4761	7.9169	8.3930
5.5	34.589	0.001006	25.767	25.768	144.86	2277.0	2421.9	144.87	2418.7	2563.6	0.4994	7.8598	8.3592
6.0	36.167	0.001006	23.737	23.738	151.46	2272.5	2424.0	151.47	2415.0	2566.5	0.5208	7.8075	8.3283
6.5	37.635	0.001007	22.013	22.014	157.60	2268.4	2426.0	157.61	2411.5	2569.1	0.5406	7.7594	8.3000
7.0	39.008	0.001008	20.528	20.529	163.35	2264.5	2427.9	163.35	2408.3	2571.6	0.5590	7.7148	8.2738
7.5	40.299	0.001008	19.236	19.237	168.75	2260.9	2429.6	168.76	2405.1	2573.9	0.5763	7.6731	8.2494
8.0	41.518	0.001008	18.102	18.103	173.85	2257.5	2431.3	173.85	2402.3	2576.1	0.5925	7.6342	8.2267
8.5	42.673	0.001009	17.098	17.099	178.68	2254.1	2432.8	178.68	2399.4	2578.1	0.6078	7.5975	8.2053
9.0	43.771	0.001009	16.202	16.203	183.27	2251.0	2434.3	183.27	2396.8	2580.1	0.6223	7.5629	8.1852
9.5	44.817	0.001010	15.398	15.399	187.64	2248.1	2435.7	187.65	2394.4	2582.0	0.6361	7.5301	8.1662
10	45.817	0.001010	14.673	14.674	191.82	2245.2	2437.0	191.83	2392.0	2583.8	0.6493	7.4989	8.1482
15	53.983	0.001014	10.022	10.023	225.97	2221.9	2447.9	225.98	2372.2	2598.2	0.7550	7.2516	8.0066
20	60.073	0.001017	7.6489	7.6499	251.44	2204.5	2455.9	251.46	2357.4	2608.9	0.8321	7.0747	7.9068

P kPa	T °C	v_l m ³ /kg	v_{lg} m ³ /kg	v_g m ³ /kg	u_l kJ/kg	u_{lg} kJ/kg	u_g kJ/kg	h_l kJ/kg	h_{lg} kJ/kg	h_g kJ/kg	s_l kJ/kg.K	s_{lg} kJ/kg.K	s_g kJ/kg.K
25	64.980	0.001020	6.2038	6.2048	271.97	2190.3	2462.3	271.99	2345.4	2617.4	0.8933	6.9365	7.8298
30	69.114	0.001022	5.2288	5.2298	289.27	2178.4	2467.7	289.30	2335.3	2624.6	0.9441	6.8231	7.7672
35	72.700	0.001024	4.5252	4.5262	304.28	2168.0	2472.3	304.32	2326.4	2630.7	0.9878	6.7266	7.7144
40	75.877	0.001026	3.9930	3.9940	317.59	2158.8	2476.4	317.64	2318.5	2636.1	1.0261	6.6427	7.6688
45	78.736	0.001028	3.5759	3.5769	329.58	2150.4	2480.0	329.62	2311.3	2640.9	1.0603	6.5684	7.6287
50	81.339	0.001030	3.2398	3.2408	340.49	2142.8	2483.3	340.54	2304.8	2645.3	1.0912	6.5016	7.5928
60	85.949	0.001033	2.7314	2.7324	359.84	2129.2	2489.0	359.90	2293.1	2653.0	1.1454	6.3856	7.5310
70	89.956	0.001036	2.3644	2.3654	376.68	2117.3	2494.0	376.75	2282.9	2659.6	1.1920	6.2869	7.4789
80	93.511	0.001038	2.0866	2.0876	391.63	2106.7	2498.3	391.71	2273.6	2665.3	1.2330	6.2009	7.4339
90	96.713	0.001041	1.8688	1.8698	405.11	2097.1	2502.2	405.20	2265.3	2670.5	1.2696	6.1247	7.3943
100	99.632	0.001043	1.6933	1.6943	417.41	2088.3	2505.7	417.51	2257.6	2675.1	1.3027	6.0562	7.3589
101.32	100.00	0.001043	1.6727	1.6737	418.96	2087.1	2506.1	419.06	2256.6	2675.7	1.3069	6.0476	7.3545
125	105.99	0.001048	1.3742	1.3752	444.25	2068.9	2513.2	444.38	2240.7	2685.1	1.3741	5.9100	7.2841
150	111.38	0.001053	1.1584	1.1595	467.02	2052.4	2519.4	467.18	2226.2	2693.4	1.4338	5.7894	7.2232
175	116.07	0.001057	1.0027	1.0038	486.89	2037.8	2524.7	487.08	2213.3	2700.4	1.4851	5.6866	7.1717
200	120.24	0.001060	0.8848	0.8859	504.59	2024.8	2529.4	504.80	2201.7	2706.5	1.5304	5.5968	7.1272
225	124.01	0.001064	0.7923	0.7934	520.59	2012.9	2533.5	520.83	2191.2	2712.0	1.5708	5.5172	7.0880
250	127.44	0.001067	0.7177	0.7188	535.22	2001.9	2537.1	535.49	2181.3	2716.8	1.6075	5.4454	7.0529
275	130.61	0.001070	0.6563	0.6574	548.73	1991.8	2540.5	549.02	2172.3	2721.3	1.6411	5.3800	7.0211
300	133.56	0.001073	0.6048	0.6059	561.29	1982.2	2543.5	561.61	2163.7	2725.3	1.6721	5.3200	6.9921
325	136.31	0.001076	0.5609	0.5620	573.04	1973.3	2546.3	573.39	2155.6	2729.0	1.7009	5.2645	6.9654
350	138.89	0.001079	0.5232	0.5243	584.10	1964.8	2548.9	584.48	2147.9	2732.4	1.7278	5.2129	6.9407
375	141.33	0.001081	0.4903	0.4914	594.56	1956.7	2551.3	594.96	2140.6	2735.6	1.7531	5.1646	6.9177

P kPa	T °C	v_l m ³ /kg	v_{lg} m ³ /kg	v_g m ³ /kg	u_l kJ/kg	u_{lg} kJ/kg	u_g kJ/kg	h_l kJ/kg	h_{lg} kJ/kg	h_g kJ/kg	s_l kJ/kg.K	s_{lg} kJ/kg.K	s_g kJ/kg.K
400	143.64	0.001084	0.4614	0.4625	604.47	1949.0	2553.5	604.91	2133.6	2738.5	1.7770	5.1191	6.8961
425	145.84	0.001086	0.4357	0.4368	613.91	1941.7	2555.6	614.37	2126.9	2741.3	1.7996	5.0762	6.8758
450	147.94	0.001088	0.4129	0.4140	622.93	1934.7	2557.6	623.42	2120.5	2743.9	1.8211	5.0356	6.8567
475	149.94	0.001090	0.3923	0.3934	631.56	1927.8	2559.4	632.07	2114.2	2746.3	1.8415	4.9971	6.8386
500	151.87	0.001093	0.3738	0.3749	639.84	1921.4	2561.2	640.38	2108.2	2748.6	1.8610	4.9604	6.8214
550	155.49	0.001097	0.3415	0.3426	655.48	1908.9	2564.4	656.08	2096.8	2752.9	1.8977	4.8917	6.7894
600	158.86	0.001101	0.3145	0.3156	670.05	1897.3	2567.3	670.71	2086.0	2756.7	1.9315	4.8286	6.7601
650	162.02	0.001104	0.2915	0.2926	683.71	1886.2	2569.9	684.42	2075.8	2760.2	1.9631	4.7699	6.7330
700	164.98	0.001108	0.2717	0.2728	696.58	1875.8	2572.4	697.35	2066.0	2763.3	1.9925	4.7154	6.7079
750	167.79	0.001111	0.2544	0.2555	708.76	1865.8	2574.6	709.59	2056.6	2766.2	2.0203	4.6642	6.6845
800	170.44	0.001115	0.2393	0.2404	720.33	1856.3	2576.6	721.23	2047.7	2768.9	2.0464	4.6161	6.6625
850	172.97	0.001118	0.2258	0.2269	731.37	1847.1	2578.5	732.32	2039.1	2771.4	2.0712	4.5706	6.6418
900	175.39	0.001121	0.2138	0.2149	741.92	1838.3	2580.2	742.93	2030.7	2773.6	2.0948	4.5274	6.6222
950	177.70	0.001124	0.2030	0.2041	752.03	1829.8	2581.8	753.10	2022.6	2775.7	2.1173	4.4863	6.6036
1000	179.92	0.001127	0.1933	0.1944	761.75	1821.6	2583.3	762.88	2014.8	2777.7	2.1388	4.4471	6.5859
1100	184.10	0.001133	0.1764	0.1775	780.14	1805.9	2586.0	781.38	1999.8	2781.2	2.1793	4.3736	6.5529
1200	188.00	0.001138	0.1622	0.1633	797.31	1791.1	2588.4	798.68	1985.6	2784.3	2.2167	4.3059	6.5226
1300	191.64	0.001144	0.1501	0.1512	813.44	1777.1	2590.5	814.93	1972.1	2787.0	2.2515	4.2430	6.4945
1400	195.08	0.001149	0.1397	0.1408	828.67	1763.6	2592.3	830.28	1959.1	2789.4	2.2842	4.1841	6.4683
1500	198.33	0.001154	0.1305	0.1317	843.12	1750.8	2593.9	844.85	1946.7	2791.5	2.3150	4.1288	6.4438
1600	201.41	0.001159	0.1225	0.1237	856.88	1738.4	2595.3	858.73	1934.6	2793.3	2.3441	4.0766	6.4207
1700	204.35	0.001163	0.1155	0.1167	870.02	1726.6	2596.6	872.00	1923.0	2795.0	2.3717	4.0272	6.3989

P kPa	T °C	v_l m ³ /kg	v_{lg} m ³ /kg	v_g m ³ /kg	u_l kJ/kg	u_{lg} kJ/kg	u_g kJ/kg	h_l kJ/kg	h_{lg} kJ/kg	h_g kJ/kg	s_l kJ/kg.K	s_{lg} kJ/kg.K	s_g kJ/kg.K
1800	207.15	0.001168	0.1092	0.1104	882.61	1715.1	2597.7	884.71	1911.7	2796.4	2.3980	3.9801	6.3781
1900	209.84	0.001172	0.1035	0.1047	894.70	1704.0	2598.7	896.92	1900.7	2797.6	2.4231	3.9353	6.3584
2000	212.42	0.001177	0.09841	0.09959	906.33	1693.2	2599.5	908.69	1890.0	2798.7	2.4471	3.8925	6.3396
2250	218.45	0.001187	0.08753	0.08872	933.70	1667.5	2601.2	936.37	1864.4	2800.8	2.5032	3.7926	6.2958
2500	223.99	0.001197	0.07875	0.07995	958.98	1643.3	2602.3	961.97	1840.2	2802.2	2.5544	3.7016	6.2560
2750	229.11	0.001207	0.07151	0.07272	982.53	1620.5	2603.0	985.85	1817.2	2803.0	2.6016	3.6178	6.2194
3000	233.89	0.001217	0.06544	0.06666	1004.6	1598.7	2603.3	1008.3	1795.0	2803.3	2.6454	3.5401	6.1855
3250	238.37	0.001226	0.06027	0.06150	1025.5	1577.7	2603.2	1029.5	1773.6	2803.1	2.6865	3.4673	6.1538
3500	242.60	0.001235	0.05582	0.05705	1045.3	1557.6	2602.9	1049.6	1753.0	2802.6	2.7251	3.3989	6.1240
3750	246.59	0.001244	0.05194	0.05318	1064.2	1538.1	2602.3	1068.8	1732.9	2801.7	2.7616	3.3341	6.0957
4000	250.39	0.001252	0.04852	0.04977	1082.2	1519.3	2601.5	1087.2	1713.4	2800.6	2.7962	3.2727	6.0689
5000	263.98	0.001286	0.03815	0.03944	1147.8	1448.7	2596.5	1154.2	1639.5	2793.7	2.9201	3.0524	5.9725
6000	275.62	0.001319	0.03112	0.03244	1205.4	1383.9	2589.3	1213.3	1570.6	2783.9	3.0266	2.8620	5.8886
7000	285.86	0.001352	0.02602	0.02737	1257.5	1322.7	2580.2	1267.0	1504.8	2771.8	3.1211	2.6919	5.8130
8000	295.04	0.001384	0.02214	0.02352	1305.5	1264.1	2569.6	1316.6	1441.2	2757.8	3.2066	2.5365	5.7431
9000	303.38	0.001418	0.01906	0.02048	1350.3	1207.3	2557.6	1363.1	1378.9	2742.0	3.2855	2.3916	5.6771
10,000	311.03	0.001452	0.01658	0.01803	1392.8	1151.4	2544.2	1407.3	1317.2	2724.5	3.3591	2.2548	5.6139
11,000	318.11	0.001488	0.01450	0.01599	1433.3	1096.2	2529.5	1449.7	1255.7	2705.4	3.4287	2.1238	5.5525
12,000	324.71	0.001526	0.01273	0.01426	1472.4	1041.0	2513.4	1490.7	1193.8	2684.5	3.4953	1.9968	5.4921
13,000	330.89	0.001566	0.01121	0.01278	1510.5	985.20	2495.7	1530.9	1130.9	2661.8	3.5595	1.8723	5.4318
14,000	336.70	0.001610	0.009870	0.01148	1547.9	928.40	2476.3	1570.4	1066.7	2637.1	3.6220	1.7491	5.3711

P kPa	T °C	v _l m ³ /kg	v _{lg} m ³ /kg	v _g m ³ /kg	u _l kJ/kg	u _{lg} kJ/kg	u _g kJ/kg	h _l kJ/kg	h _{lg} kJ/kg	h _g kJ/kg	s _l kJ/kg.K	s _{lg} kJ/kg.K	s _g kJ/kg.K
15,000	342.19	0.001657	0.008683	0.01034	1585.0	870.00	2455.0	1609.8	1000.3	2610.1	3.6837	1.6255	5.3092
16,000	347.39	0.001710	0.007600	0.009310	1622.1	809.20	2431.3	1649.5	930.80	2580.3	3.7452	1.4999	5.2451
17,000	352.34	0.001770	0.006603	0.008373	1659.9	744.90	2404.8	1690.0	857.10	2547.1	3.8073	1.3704	5.1777
18,000	357.04	0.001840	0.005665	0.007505	1698.9	675.70	2374.6	1732.0	777.70	2509.7	3.8714	1.2340	5.1054
19,000	361.52	0.001925	0.004756	0.006681	1740.3	599.00	2339.3	1776.8	689.40	2466.2	3.9393	1.0862	5.0255
20,000	365.80	0.002036	0.003838	0.005874	1786.0	510.10	2296.1	1826.7	586.90	2413.6	4.0146	0.9184	4.9330
21,000	369.88	0.002200	0.002820	0.005020	1841.4	396.00	2237.4	1887.6	455.20	2342.8	4.1062	0.7079	4.8141
22,000	373.77	0.002702	0.000952	0.003654	1953.4	142.80	2096.2	2012.8	163.60	2176.5	4.2866	0.2530	4.5486
22,055	373.98	0.00311	-	0.00311	2017	-	2017	2086	-	2086	4.409	-	4.409

Table A 2.2: Properties of SATURATED WATER – Temperature Table

T °C	P kPa	v_l m ³ /kg	v_{lg} m ³ /kg	v_g m ³ /kg	u_l kJ/kg	u_{lg} kJ/kg	u_g kJ/kg	h_l kJ/kg	h_{lg} kJ/kg	h_g kJ/kg	s_l kJ/kg.K	s_{lg} kJ/kg.K	s_g kJ/kg.K
5	0.8726	0.001000	147.02	147.02	21.020	2360.4	2381.4	21.021	2488.7	2509.7	0.07626	8.9473	9.0236
10	1.2281	0.001000	106.32	106.32	41.986	2346.3	2388.3	41.988	2476.9	2518.9	0.1510	8.7476	8.8986
15	1.7056	0.001001	77.896	77.897	62.915	2332.3	2395.2	62.917	2465.1	2528.0	0.2242	8.5550	8.7792
20	2.3388	0.001002	57.777	57.778	83.833	2318.2	2402.0	83.835	2453.4	2537.2	0.2962	8.3689	8.6651
25	3.1690	0.001003	43.356	43.357	104.75	2304.1	2408.9	104.75	2441.6	2546.3	0.3670	8.1888	8.5558
30	4.2455	0.001004	32.895	32.896	125.67	2290.0	2415.7	125.67	2429.6	2555.3	0.4365	8.0148	8.4513
35	5.6267	0.001006	25.219	25.220	146.58	2275.9	2422.5	146.59	2417.8	2564.4	0.5050	7.8461	8.3511
40	7.3814	0.001008	19.527	19.528	167.50	2261.7	2429.2	167.50	2405.9	2573.4	0.5723	7.6827	8.2550
45	9.5898	0.001010	15.262	15.263	188.41	2247.5	2435.9	188.42	2393.9	2582.3	0.6385	7.5244	8.1629
50	12.344	0.001012	12.036	12.037	209.31	2233.3	2442.6	209.33	2381.9	2591.2	0.7037	7.3708	8.0745
55	15.752	0.001015	9.5716	9.5726	230.22	2219.0	2449.2	230.24	2369.8	2600.0	0.7679	7.2217	7.9896
60	19.932	0.001017	7.6733	7.6743	251.13	2204.7	2455.8	251.15	2357.7	2608.8	0.8312	7.0768	7.9080
65	25.022	0.001020	6.1986	6.1996	272.05	2190.3	2462.4	272.08	2345.4	2617.5	0.8935	6.9360	7.8295
70	31.176	0.001023	5.0437	5.0447	292.98	2175.8	2468.8	293.01	2333.1	2626.1	0.9549	6.7991	7.7540
75	38.563	0.001026	4.1323	4.1333	313.92	2161.3	2475.2	313.96	2320.6	2634.6	1.0155	6.6658	7.6813
80	47.373	0.001029	3.4078	3.4088	334.88	2146.7	2481.6	334.93	2308.2	2643.1	1.0753	6.5359	7.6112
85	57.815	0.001032	2.8279	2.8289	355.86	2132.0	2487.9	355.92	2295.5	2651.4	1.1343	6.4093	7.5436
90	70.117	0.001036	2.3607	2.3617	376.86	2117.1	2494.0	376.93	2282.7	2659.6	1.1925	6.2859	7.4784
95	84.529	0.001040	1.9818	1.9828	397.89	2102.2	2500.1	397.98	2269.7	2667.7	1.2501	6.1653	7.4154
100	101.32	0.001043	1.6726	1.6736	418.96	2087.1	2506.1	419.06	2256.6	2675.7	1.3069	6.0476	7.3545
105	120.79	0.001047	1.4190	1.4200	440.05	2072.1	2512.1	440.18	2243.4	2683.6	1.3630	5.9326	7.2956
110	143.24	0.001052	1.2095	1.2106	461.19	2056.7	2517.9	461.34	2230.0	2691.3	1.4186	5.8200	7.2386
115	169.02	0.001056	1.0359	1.0370	482.36	2041.1	2523.5	482.54	2216.3	2698.8	1.4735	5.7098	7.1833
120	198.48	0.001060	0.8911	0.8922	503.57	2025.5	2529.1	503.78	2202.4	2706.2	1.5278	5.6019	7.1297
125	232.01	0.001065	0.7698	0.7709	524.82	2009.7	2534.5	525.07	2188.3	2713.4	1.5815	5.4962	7.0777

T °C	P kPa	v_l m ³ /kg	v_{lg} m ³ /kg	v_g m ³ /kg	u_l kJ/kg	u_{lg} kJ/kg	u_g kJ/kg	h_l kJ/kg	h_{lg} kJ/kg	h_g kJ/kg	s_l kJ/kg.K	s_{lg} kJ/kg.K	s_g kJ/kg.K
130	270.02	0.001070	0.6676	0.6687	546.12	1993.7	2539.8	546.41	2174.0	2720.4	1.6346	5.3926	7.0272
135	312.93	0.001075	0.5813	0.5824	567.46	1977.5	2545.0	567.80	2159.4	2727.2	1.6873	5.2907	6.9780
140	361.19	0.001080	0.5079	0.5090	588.85	1961.2	2550.0	589.24	2144.6	2733.8	1.7394	5.1908	6.9302
145	415.29	0.001085	0.4453	0.4464	610.30	1944.5	2554.8	610.75	2129.4	2740.2	1.7910	5.0926	6.8836
150	475.72	0.001090	0.3918	0.3929	631.80	1927.7	2559.5	632.32	2114.1	2746.4	1.8421	4.9960	6.8381
155	542.99	0.001096	0.3457	0.3468	653.35	1910.7	2564.0	653.95	2098.4	2752.3	1.8927	4.9010	6.7937
160	617.66	0.001102	0.3060	0.3071	674.97	1893.3	2568.3	675.65	2082.3	2758.0	1.9429	4.8074	6.7503
165	700.29	0.001108	0.2716	0.2727	696.65	1875.7	2572.4	697.43	2065.9	2763.3	1.9927	4.7151	6.7078
170	791.47	0.001114	0.2417	0.2428	718.40	1857.9	2576.3	719.28	2049.2	2768.5	2.0421	4.6241	6.6662
175	891.80	0.001121	0.2157	0.2168	740.22	1839.7	2579.9	741.22	2032.1	2773.3	2.0910	4.5344	6.6254
180	1001.9	0.001127	0.1929	0.1940	762.12	1821.3	2583.4	763.25	2014.6	2777.8	2.1397	4.4456	6.5853
185	1122.5	0.001134	0.1730	0.1741	784.10	1802.5	2586.6	785.37	1996.6	2782.0	2.1879	4.3580	6.5459
190	1254.2	0.001141	0.1554	0.1565	806.17	1783.4	2589.6	807.60	1978.2	2785.8	2.2358	4.2713	6.5071
195	1397.6	0.001149	0.1399	0.1410	828.33	1764.0	2592.3	829.93	1959.5	2789.4	2.2834	4.1855	6.4689
200	1553.6	0.001156	0.1261	0.1273	850.58	1744.1	2594.7	852.38	1940.1	2792.5	2.3308	4.1004	6.4312
205	1722.9	0.001164	0.1140	0.1152	872.95	1723.9	2596.9	874.95	1920.4	2795.3	2.3778	4.0162	6.3940
210	1906.2	0.001173	0.1032	0.1044	895.43	1703.3	2598.7	897.66	1900.0	2797.7	2.4246	3.9326	6.3572
215	2104.2	0.001181	0.09357	0.09475	918.02	1682.3	2600.3	920.51	1879.2	2799.7	2.4712	3.8496	6.3208
220	2317.8	0.001190	0.08497	0.08616	940.75	1660.9	2601.6	943.51	1857.8	2801.3	2.5175	3.7672	6.2847
225	2547.9	0.001199	0.07726	0.07846	963.61	1638.9	2602.5	966.67	1835.7	2802.4	2.5637	3.6851	6.2488
230	2795.1	0.001209	0.07034	0.07155	986.62	1616.5	2603.1	990.00	1813.1	2803.1	2.6097	3.6034	6.2131
235	3060.4	0.001219	0.06412	0.06534	1009.8	1593.5	2603.3	1013.5	1789.8	2803.3	2.6556	3.5221	6.1777
240	3344.7	0.001229	0.05851	0.05974	1033.1	1570.0	2603.1	1037.2	1765.8	2803.0	2.7013	3.4410	6.1423
245	3648.8	0.001240	0.05345	0.05469	1056.6	1546.0	2602.6	1061.2	1740.9	2802.1	2.7470	3.3600	6.1070
250	3973.6	0.001251	0.04886	0.05011	1080.3	1521.3	2601.6	1085.3	1715.4	2800.7	2.7926	3.2791	6.0717
255	4320.2	0.001263	0.04470	0.04596	1104.3	1495.9	2600.2	1109.7	1689.1	2798.8	2.8382	3.1981	6.0363
260	4689.4	0.001276	0.04091	0.04219	1128.4	1470.0	2598.4	1134.4	1661.8	2796.2	2.8838	3.1171	6.0009

T °C	P kPa	v_l m ³ /kg	v_{lg} m ³ /kg	v_g m ³ /kg	u_l kJ/kg	u_{lg} kJ/kg	u_g kJ/kg	h_l kJ/kg	h_{lg} kJ/kg	h_g kJ/kg	s_l kJ/kg.K	s_{lg} kJ/kg.K	s_g kJ/kg.K
265	5082.3	0.001289	0.03747	0.03876	1152.8	1443.2	2596.0	1159.3	1633.7	2793.0	2.9294	3.0358	5.9652
270	5499.9	0.001303	0.03434	0.03564	1177.4	1415.8	2593.2	1184.6	1604.5	2789.1	2.9751	2.9542	5.9293
275	5943.1	0.001317	0.03146	0.03278	1202.3	1387.4	2589.7	1210.1	1574.4	2784.5	3.0209	2.8722	5.8931
280	6413.2	0.001332	0.02883	0.03016	1227.5	1358.2	2585.7	1236.1	1543.1	2779.2	3.0669	2.7896	5.8565
285	6911.1	0.001349	0.02642	0.02777	1253.1	1328.0	2581.1	1262.4	1510.6	2773.0	3.1131	2.7064	5.8195
290	7438.0	0.001366	0.02419	0.02556	1279.0	1296.7	2575.7	1289.1	1476.8	2765.9	3.1595	2.6223	5.7818
295	7995.2	0.001384	0.02216	0.02354	1305.3	1264.4	2569.7	1316.3	1441.5	2757.8	3.2062	2.5372	5.7454
300	8583.8	0.001404	0.02027	0.02167	1332.0	1230.8	2562.8	1344.1	1404.6	2748.7	3.2534	2.4569	5.7042
305	9205.1	0.001425	0.01852	0.01994	1359.2	1195.8	2555.0	1372.3	1366.2	2738.5	3.3010	2.3630	5.6640
310	9860.5	0.001447	0.01689	0.01834	1387.0	1159.2	2546.2	1401.2	1325.8	2727.0	3.3491	2.2735	5.6226
315	10,550	0.001472	0.01539	0.01686	1415.3	1121.0	2536.3	1430.8	1283.4	2714.2	3.3979	2.1820	5.5799
320	11,280	0.001498	0.01398	0.01548	1444.4	1080.8	2525.2	1461.3	1238.4	2699.7	3.4476	2.0880	5.5356
325	12,050	0.001528	0.01266	0.01419	1474.2	1038.4	2512.6	1492.6	1190.9	2683.5	3.4983	1.9910	5.4893
330	12,850	0.001560	0.01142	0.01298	1504.9	993.50	2498.4	1525.0	1140.3	2665.3	3.5501	1.8906	5.4407
335	13,700	0.001596	0.01025	0.01185	1536.8	945.50	2482.3	1558.6	1086.1	2644.7	3.6035	1.7854	5.3894
340	14,590	0.001637	0.009153	0.01079	1569.9	894.00	2463.9	1593.8	1027.5	2621.3	3.6587	1.6758	5.3345
345	15,530	0.001684	0.008094	0.009778	1604.7	838.00	2442.7	1630.9	963.60	2594.5	3.7164	1.5589	5.2753
350	16,520	0.001740	0.007072	0.008812	1641.7	776.20	2417.9	1670.4	893.10	2563.5	3.7774	1.4331	5.2105
355	17,560	0.001808	0.006071	0.007879	1681.5	706.90	2388.4	1713.3	813.40	2526.7	3.8429	1.2950	5.1379
360	18,660	0.001894	0.005068	0.006962	1725.6	626.60	2352.2	1761.0	721.00	2482.0	3.9153	1.1389	5.0542
365	19,810	0.002012	0.004017	0.006029	1776.8	528.40	2305.2	1816.7	607.90	2424.6	3.9994	0.9526	4.9520
370	21,030	0.002207	0.002786	0.004993	1843.3	391.90	2235.2	1889.7	450.50	2340.2	4.1094	0.7004	4.8098
373	22,055	0.00311	-	0.00311	2017	-	2017	2086	-	2086	4.409	-	4.409

Table A2.3: Properties of WATER in Compressed Liquid State

P kPa	T °C	v m ³ /kg	u kJ/kg	h kJ/kg	s kJ/kg.K
5,000	10	0.000998	41.860	46.850	0.1505
	30	0.001002	125.21	130.22	0.4350
	50	0.001010	208.58	213.63	0.7014
	70	0.001020	291.97	297.07	0.9520
	90	0.001034	375.59	380.75	1.1890
	110	0.001049	459.63	464.88	1.4145
	130	0.001067	544.28	549.61	1.6301
	150	0.001087	629.68	635.12	1.8371
	170	0.001111	716.04	721.59	2.0367
	190	0.001138	803.64	809.33	2.2304
	210	0.001169	892.89	898.74	2.4193
	230	0.001206	984.40	990.43	2.6053
	250	0.001250	1079.0	1085.3	2.7901

P kPa	T °C	v m ³ /kg	u kJ/kg	h kJ/kg	s kJ/kg.K
10,000	10	0.000996	41.729	51.685	0.1500
	30	0.001000	124.75	134.75	0.4334
	50	0.001008	207.85	217.93	0.6991
	70	0.001018	290.98	301.16	0.9491
	90	0.001031	374.32	384.63	1.1855
	110	0.001046	458.07	468.53	1.4104
	130	0.001064	542.38	553.02	1.6253
	150	0.001084	627.41	638.25	1.8316
	170	0.001107	713.31	724.39	2.0305
	190	0.001133	800.37	811.70	2.2232
	210	0.001164	888.94	900.57	2.4111
	230	0.001199	979.55	991.54	2.5955
	250	0.001241	1073.0	1085.4	2.7784
	270	0.001292	1170.3	1183.2	2.9619
	290	0.001357	1273.5	1287.1	3.1497

P kPa	T °C	v m ³ /kg	u kJ/kg	h kJ/kg	s kJ/kg.K
15,000	10	0.000993	41.593	56.493	0.1494
	30	0.000998	124.30	139.27	0.4319
	50	0.001006	207.13	222.22	0.6968
	70	0.001016	290.01	305.25	0.9461
	90	0.001029	373.08	388.51	1.1820
	110	0.001044	456.54	472.20	1.4063
	130	0.001061	540.53	556.45	1.6206
	150	0.001081	625.19	641.41	1.8263
	170	0.001104	710.67	727.22	2.0244
	190	0.001129	797.20	814.14	2.2163
	210	0.001159	885.13	902.51	2.4030
	230	0.001193	974.91	992.80	2.5862
	250	0.001233	1067.2	1085.7	2.7672
	270	0.001281	1163.0	1182.2	2.9481
	290	0.001341	1263.7	1283.8	3.1319
	310	0.001421	1372.1	1393.5	3.3231
	330	0.001539	1494.7	1517.8	3.5327

P kPa	T °C	v m ³ /kg	u kJ/kg	h kJ/kg	s kJ/kg.K
20,000	10	0.000991	41.453	61.274	0.1488
	30	0.000996	123.86	143.77	0.4303
	50	0.001003	206.43	226.50	0.6946
	70	0.001014	289.05	309.33	0.9432
	90	0.001026	371.86	392.39	1.1785
	110	0.001041	455.04	475.87	1.4023
	130	0.001058	538.73	559.90	1.6160
	150	0.001078	623.03	644.59	1.8211
	170	0.001100	708.10	730.10	2.0185
	190	0.001125	794.14	816.64	2.2095
	210	0.001154	881.46	904.53	2.3953
	230	0.001187	970.48	994.21	2.5771
	250	0.001225	1061.8	1086.3	2.7565
	270	0.001271	1156.1	1181.5	2.9352
	290	0.001327	1254.7	1281.3	3.1155
	310	0.001399	1359.7	1387.7	3.3011
	330	0.001499	1475.0	1505.0	3.4988
	350	0.001665	1612.1	1645.4	3.7277

P kPa	T °C	v m ³ /kg	u kJ/kg	h kJ/kg	s kJ/kg.K
30,000	10	0.000987	41.158	70.757	0.1473
	30	0.000992	122.99	152.73	0.4271
	50	0.000999	205.07	235.05	0.6900
	70	0.001010	287.20	317.49	0.9375
	90	0.001022	369.51	400.16	1.1717
	110	0.001036	452.15	483.24	1.3944
	130	0.001053	535.24	566.83	1.6070
	150	0.001072	618.87	651.03	1.8109
	170	0.001093	703.17	735.96	2.0070
	190	0.001117	788.29	821.81	2.1965
	210	0.001144	874.50	908.83	2.3804
	230	0.001175	962.13	997.39	2.5600
	250	0.001211	1051.6	1088.0	2.7365
	270	0.001253	1143.6	1181.2	2.9113
	290	0.001303	1238.8	1277.9	3.0862
	310	0.001364	1338.6	1379.5	3.2635
	330	0.001443	1444.9	1488.1	3.4466
	350	0.001552	1561.5	1608.1	3.6421
	370	0.001726	1697.8	1749.6	3.8656

P kPa	T °C	v m ³ /kg	u kJ/kg	h kJ/kg	s kJ/kg.K
50,000	10	0.000978	40.519	89.428	0.1439
	30	0.000984	121.31	170.48	0.4205
	50	0.000991	202.46	252.03	0.6810
	70	0.001001	283.69	333.76	0.9264
	90	0.001013	365.06	415.73	1.1585
	110	0.001027	446.70	498.06	1.3792
	130	0.001043	528.70	580.85	1.5898
	150	0.001061	611.12	664.16	1.7915
	170	0.001081	694.04	748.07	1.9853
	190	0.001103	777.56	832.70	2.1720
	210	0.001128	861.87	918.25	2.3529
	230	0.001155	947.19	1005.0	2.5287
	250	0.001187	1033.8	1093.2	2.7006
	270	0.001223	1122.1	1183.3	2.8696
	290	0.001264	1212.5	1275.8	3.0368
	310	0.001313	1305.6	1371.3	3.2035
	330	0.001371	1402.2	1470.7	3.3712
	350	0.001442	1503.2	1575.3	3.5417
	370	0.001533	1610.2	1686.9	3.7179

Table A2.4: Properties of Superheated Steam

P kPa	T °C	v m³/kg	u kJ/kg	h kJ/kg	s kJ/kg.K
10	(45.82)	(14.674)	(2437.0)	(2583.8)	(8.1482)
	50	14.869	2443.1	2591.8	8.1731
	100	17.196	2515.0	2687.0	8.4471
	150	19.513	2587.4	2782.5	8.6873
	200	21.826	2660.8	2879.0	8.9030
	250	24.136	2735.5	2976.9	9.0995
	300	26.446	2811.7	3076.2	9.2808
	350	28.755	2889.5	3177.0	9.4494
	400	31.063	2968.8	3279.4	9.6075
	450	33.372	3049.7	3383.4	9.7565
	500	35.680	3132.4	3489.2	9.8979
	550	37.988	3216.7	3596.6	10.032
	600	40.296	3302.8	3705.7	10.161
	650	42.604	3390.6	3816.7	10.285
	700	44.912	3480.2	3929.4	10.404
	750	47.220	3571.6	4043.8	10.518
	800	49.527	3664.8	4160.1	10.629
	850	51.835	3759.7	4278.1	10.737

P kPa	T °C	v m³/kg	u kJ/kg	h kJ/kg	s kJ/kg.K
50	(81.33)	(3.2408)	(2483.3)	(2645.3)	(7.5928)
	100	3.4188	2511.2	2682.1	7.6941
	150	3.8895	2585.2	2779.7	7.9394
	200	4.3560	2659.4	2877.2	8.1572
	250	4.8205	2734.5	2975.6	8.3548
	300	5.2840	2811.0	3075.2	8.5367
	350	5.7469	2888.9	3176.2	8.7057
	400	6.2094	2968.3	3278.8	8.8640
	450	6.6717	3049.3	3382.9	9.0132
	500	7.1338	3132.0	3488.7	9.1547
	550	7.5958	3216.4	3596.2	9.2894
	600	8.0577	3302.5	3705.4	9.4182
	650	8.5195	3390.4	3816.4	9.5417
	700	8.9813	3480.1	3929.1	9.6606
	750	9.4430	3571.5	4043.6	9.7754
	800	9.9047	3664.7	4159.9	9.8863
	850	10.366	3759.6	4277.9	9.9938

P kPa	T °C	v m³/kg	u kJ/kg	h kJ/kg	s kJ/kg.K
100	(99.63)	(1.6943)	(2505.7)	(2675.1)	(7.3589)
	100	1.6961	2506.3	2675.9	7.3609
	150	1.9364	2582.4	2776.1	7.6129
	200	2.1723	2657.6	2874.8	7.8335
	250	2.4061	2733.3	2973.9	8.0325
	300	2.6388	2810.1	3073.9	8.2152
	350	2.8709	2888.2	3175.3	8.3846
	400	3.1027	2967.7	3278.0	8.5432
	450	3.3342	3048.9	3382.3	8.6927
	500	3.5655	3131.6	3488.2	8.8342
	550	3.7968	3216.1	3595.8	8.9690
	600	4.0279	3302.3	3705.0	9.0979
	650	4.2590	3390.2	3816.1	9.2216
	700	4.4900	3479.8	3928.8	9.3405
	750	4.7210	3571.3	4043.4	9.4553
	800	4.9519	3664.5	4159.7	9.5662
	850	5.1828	3759.4	4277.7	9.6738

P kPa	T °C	v m³/kg	u kJ/kg	h kJ/kg	s kJ/kg.K
101.33	(100.00)	(1.6737)	(2506.1)	(2675.7)	(7.3545)
	150	1.9108	2582.3	2776.0	7.6066
	200	2.1436	2657.5	2874.8	7.8273
	250	2.3744	2733.3	2973.8	8.0264
	300	2.6041	2810.0	3073.9	8.2090
	350	2.8332	2888.1	3175.2	8.3785
	400	3.0619	2967.7	3278.0	8.5371
	450	3.2904	3048.9	3382.3	8.6865
	500	3.5187	3131.6	3488.2	8.8281
	550	3.7469	3216.1	3595.8	8.9629
	600	3.9750	3302.2	3705.0	9.0918
	650	4.2030	3390.2	3816.1	9.2154
	700	4.4310	3479.8	3928.8	9.3344
	750	4.6590	3571.3	4043.4	9.4492
	800	4.8869	3664.5	4159.7	9.5601
	850	5.1148	3759.4	4277.7	9.6677

P kPa	T °C	v m ³ /kg	u kJ/kg	h kJ/kg	s kJ/kg.K
200	(120.24)	(0.8859)	(2529.4)	(2706.5)	(7.1272)
	150	0.9597	2576.7	2768.6	7.2793
	200	1.0803	2653.9	2870.0	7.5059
	250	1.1988	2730.8	2970.5	7.7078
	300	1.3162	2808.2	3071.4	7.8920
	350	1.4329	2886.7	3173.3	8.0624
	400	1.5493	2966.6	3276.4	8.2216
	450	1.6655	3047.9	3381.0	8.3714
	500	1.7814	3130.8	3487.1	8.5133
	550	1.8973	3215.4	3594.9	8.6483
	600	2.0130	3301.7	3704.3	8.7773
	650	2.1287	3389.7	3815.4	8.9011
	700	2.2443	3479.4	3928.3	9.0201
	750	2.3599	3570.9	4042.9	9.1350
	800	2.4755	3664.1	4159.2	9.2460
	850	2.5910	3759.1	4277.3	9.3536

P kPa	T °C	v m ³ /kg	u kJ/kg	h kJ/kg	s kJ/kg.K
300	(133.56)	(0.6059)	(2543.5)	(2725.3)	(6.9921)
	150	0.6339	2570.7	2760.9	7.0779
	200	0.7163	2650.2	2865.1	7.3108
	250	0.7963	2728.2	2967.1	7.5157
	300	0.8753	2806.3	3068.9	7.7015
	350	0.9536	2885.3	3171.3	7.8729
	400	1.0315	2965.4	3274.9	8.0327
	450	1.1092	3047.0	3379.7	8.1830
	500	1.1867	3130.1	3486.1	8.3252
	550	1.2641	3214.7	3594.0	8.4604
	600	1.3414	3301.1	3703.5	8.5895
	650	1.4186	3389.1	3814.7	8.7134
	700	1.4958	3478.9	3927.7	8.8325
	750	1.5729	3570.5	4042.3	8.9475
	800	1.6500	3663.8	4158.8	9.0585
	850	1.7271	3758.8	4276.9	9.1661

P kPa	T °C	v m ³ /kg	u kJ/kg	h kJ/kg	s kJ/kg.K
400	(143.64)	(0.4625)	(2553.5)	(2738.5)	(6.8961)
	150	0.4708	2564.4	2752.8	6.9300
	200	0.5342	2646.4	2860.1	7.1699
	250	0.5951	2725.6	2963.6	7.3779
	300	0.6548	2804.4	3066.3	7.5654
	350	0.7139	2883.8	3169.4	7.7378
	400	0.7726	2964.3	3273.3	7.8982
	450	0.8311	3046.0	3378.5	8.0489
	500	0.8894	3129.3	3485.0	8.1914
	550	0.9475	3214.1	3593.1	8.3268
	600	1.0056	3300.5	3702.7	8.4561
	650	1.0636	3388.6	3814.1	8.5801
	700	1.1215	3478.5	3927.1	8.6993
	750	1.1794	3570.1	4041.8	8.8143
	800	1.2373	3663.4	4158.3	8.9254
	850	1.2951	3758.4	4276.5	9.0331

P kPa	T °C	v m ³ /kg	u kJ/kg	h kJ/kg	s kJ/kg.K
600	(158.86)	(0.3156)	(2567.3)	(2756.7)	(6.7601)
	200	0.3520	2638.5	2849.7	6.9658
	250	0.3938	2720.3	2956.6	7.1806
	300	0.4344	2800.5	3061.2	7.3716
	350	0.4742	2880.9	3165.4	7.5459
	400	0.5137	2961.9	3270.2	7.7076
	450	0.5529	3044.1	3375.9	7.8591
	500	0.5920	3127.7	3482.9	8.0022
	550	0.6309	3212.7	3591.2	8.1380
	600	0.6697	3299.3	3701.2	8.2676
	650	0.7085	3387.6	3812.7	8.3918
	700	0.7472	3477.6	3925.9	8.5112
	750	0.7859	3569.2	4040.8	8.6264
	800	0.8246	3662.7	4157.4	8.7376
	850	0.8632	3757.8	4275.7	8.8453

P kPa	T °C	v m ³ /kg	u kJ/kg	h kJ/kg	s kJ/kg.K
800	(170.44)	(0.2404)	(2576.6)	(2768.9)	(6.6625)
	200	0.2607	2630.2	2838.8	6.8151
	250	0.2931	2714.8	2949.3	7.0373
	300	0.3241	2796.6	3055.9	7.2319
	350	0.3544	2877.9	3161.4	7.4084
	400	0.3843	2959.6	3267.0	7.5713
	450	0.4139	3042.2	3373.3	7.7237
	500	0.4433	3126.1	3480.7	7.8673
	550	0.4726	3211.3	3589.4	8.0036
	600	0.5018	3298.1	3699.6	8.1335
	650	0.5310	3386.6	3811.4	8.2579
	700	0.5601	3476.7	3924.7	8.3775
	750	0.5892	3568.4	4039.8	8.4928
	800	0.6182	3661.9	4156.5	8.6041
	850	0.6472	3757.1	4274.9	8.7120

P kPa	T °C	v m ³ /kg	u kJ/kg	h kJ/kg	s kJ/kg.K
1000	(179.92)	(0.1944)	(2583.3)	(2781.2)	(6.5529)
	200	0.2059	2621.5	2827.4	6.6932
	250	0.2326	2709.2	2941.9	6.9235
	300	0.2579	2792.7	3050.6	7.1219
	350	0.2825	2874.9	3157.3	7.3005
	400	0.3066	2957.2	3263.8	7.4648
	450	0.3304	3040.3	3370.7	7.6180
	500	0.3541	3124.5	3478.6	7.7622
	550	0.3776	3210.0	3587.6	7.8989
	600	0.4011	3297.0	3698.1	8.0292
	650	0.4245	3385.5	3810.0	8.1538
	700	0.4478	3475.7	3923.6	8.2736
	750	0.4711	3567.6	4038.7	8.3890
	800	0.4944	3661.2	4155.5	8.5005
	850	0.5176	3756.4	4274.0	8.6084

P kPa	T °C	v m ³ /kg	u kJ/kg	h kJ/kg	s kJ/kg.K
1500	(198.33)	(0.1317)	(2593.9)	(2791.5)	(6.4438)
	200	0.1324	2597.5	2796.1	6.4536
	250	0.1519	2694.6	2922.4	6.7077
	300	0.1696	2782.5	3036.9	6.9168
	350	0.1866	2867.2	3147.1	7.1011
	400	0.2030	2951.2	3255.7	7.2687
	450	0.2192	3035.4	3364.2	7.4242
	500	0.2351	3120.4	3473.1	7.5699
	550	0.2510	3206.5	3583.0	7.7076
	600	0.2668	3294.0	3694.2	7.8386
	650	0.2825	3382.9	3806.6	7.9639
	700	0.2981	3473.4	3920.6	8.0841
	750	0.3137	3565.6	4036.1	8.1999
	800	0.3293	3659.3	4153.2	8.3116
	850	0.3448	3754.8	4272.0	8.4198

P kPa	T °C	v m ³ /kg	u kJ/kg	h kJ/kg	s kJ/kg.K
2500	(223.99)	(0.07995)	(2602.3)	(2802.2)	(6.2560)
	250	0.08698	2661.7	2879.1	6.4069
	300	0.09888	2760.8	3008.0	6.6424
	350	0.1097	2851.4	3125.8	6.8395
	400	0.1201	2938.9	3239.2	7.0146
	450	0.1301	3025.5	3350.9	7.1746
	500	0.1400	3112.2	3462.2	7.3235
	550	0.1497	3199.6	3573.8	7.4634
	600	0.1593	3288.0	3686.3	7.5960
	650	0.1688	3377.7	3799.8	7.7225
	700	0.1783	3468.8	3914.7	7.8436
	750	0.1878	3561.4	4030.9	7.9601
	800	0.1972	3655.6	4148.6	8.0724
	850	0.2066	3751.4	4267.9	8.1810

P kPa	T °C	v m ³ /kg	u kJ/kg	h kJ/kg	s kJ/kg.K
4000	(250.39)	(0.04977)	(2601.5)	(2800.6)	(6.0689)
	300	0.05882	2724.4	2959.7	6.3598
	350	0.06644	2826.1	3091.8	6.5811
	400	0.07340	2919.8	3213.4	6.7688
	450	0.08002	3010.3	3330.4	6.9364
	500	0.08642	3099.7	3445.4	7.0902
	550	0.09268	3189.0	3559.7	7.2335
	600	0.09884	3278.9	3674.3	7.3687
	650	0.1049	3369.8	3789.5	7.4970
	700	0.1110	3461.8	3905.7	7.6195
	750	0.1170	3555.2	4023.0	7.7371
	800	0.1229	3650.0	4141.7	7.8503
	850	0.1288	3746.3	4261.7	7.9596

P kPa	T °C	v m ³ /kg	u kJ/kg	h kJ/kg	s kJ/kg.K
6000	(275.62)	(0.03244)	(2589.3)	(2783.9)	(5.8886)
	300	0.03615	2666.3	2883.2	6.0659
	350	0.04222	2788.9	3042.2	6.3322
	400	0.04739	2892.7	3177.0	6.5404
	450	0.05214	2989.1	3301.9	6.7195
	500	0.05665	3082.4	3422.3	6.8805
	550	0.06100	3174.6	3540.6	7.0287
	600	0.06525	3266.6	3658.1	7.1673
	650	0.06942	3359.1	3775.6	7.2982
	700	0.07353	3452.4	3893.6	7.4227
	750	0.07760	3546.8	4012.4	7.5418
	800	0.08164	3642.5	4132.3	7.6561
	850	0.08565	3739.5	4253.4	7.7664

P kPa	T °C	v m ³ /kg	u kJ/kg	h kJ/kg	s kJ/kg.K
2000	(212.42)	(0.09959)	(2599.5)	(2798.7)	(6.3396)
	250	0.1114	2678.8	2901.6	6.5438
	300	0.1254	2771.8	3022.7	6.7651
	350	0.1386	2859.4	3136.6	6.9556
	400	0.1512	2945.1	3247.5	7.1269
	450	0.1635	3030.5	3357.5	7.2845
	500	0.1757	3116.3	3467.7	7.4318
	550	0.1877	3203.1	3578.4	7.5706
	600	0.1996	3291.0	3690.2	7.7024
	650	0.2114	3380.3	3803.2	7.8283
	700	0.2232	3471.1	3917.6	7.9490
	750	0.2350	3563.5	4033.5	8.0651
	800	0.2467	3657.5	4150.9	8.1771
	850	0.2584	3753.1	4269.9	8.2855

P kPa	T °C	v m ³ /kg	u kJ/kg	h kJ/kg	s kJ/kg.K
3000	(233.89)	(0.06666)	(2603.3)	(2803.3)	(6.1855)
	250	0.07056	2643.1	2854.8	6.2857
	300	0.08113	2749.2	2992.6	6.5375
	350	0.09052	2843.2	3114.8	6.7420
	400	0.09935	2932.7	3230.7	6.9210
	450	0.1079	3020.5	3344.1	7.0835
	500	0.1162	3108.1	3456.6	7.2339
	550	0.1244	3196.1	3569.1	7.3750
	600	0.1324	3285.0	3682.3	7.5084
	650	0.1404	3375.1	3796.4	7.6355
	700	0.1484	3466.5	3911.7	7.7571
	750	0.1563	3559.4	4028.3	7.8739
	800	0.1642	3653.8	4146.3	7.9865
	850	0.1720	3749.7	4265.8	8.0954

P kPa	T °C	v m ³ /kg	u kJ/kg	h kJ/kg	s kJ/kg.K
5000	(263.98)	(0.03944)	(2596.5)	(2793.7)	(5.9725)
	300	0.04530	2697.0	2923.5	6.2067
	350	0.05193	2808.0	3067.7	6.4482
	400	0.05781	2906.5	3195.5	6.6456
	450	0.06330	2999.8	3316.3	6.8187
	500	0.06856	3091.1	3433.9	6.9760
	550	0.07367	3181.8	3550.2	7.1218
	600	0.07869	3272.8	3666.2	7.2586
	650	0.08362	3364.5	3782.6	7.3882
	700	0.08850	3457.1	3899.7	7.5117
	750	0.09334	3551.0	4017.7	7.6300
	800	0.09815	3646.3	4137.0	7.7438
	850	0.1029	3742.9	4257.5	7.8536

P kPa	T °C	v m ³ /kg	u kJ/kg	h kJ/kg	s kJ/kg.K
7000	(285.86)	(0.02737)	(2580.2)	(2771.8)	(5.8130)
	300	0.02946	2631.4	2837.6	5.9293
	350	0.03523	2768.5	3015.1	6.2269
	400	0.03993	2878.4	3157.9	6.4474
	450	0.04416	2978.1	3287.3	6.6329
	500	0.04813	3073.6	3410.5	6.7978
	550	0.05194	3167.2	3530.8	6.9486
	600	0.05565	3260.3	3649.8	7.0889
	650	0.05927	3353.6	3768.5	7.2211
	700	0.06284	3447.6	3887.5	7.3466
	750	0.06636	3542.6	4007.1	7.4665
	800	0.06985	3638.7	4127.6	7.5815
	850	0.07331	3736.1	4249.2	7.6922

P kPa	T °C	v m ³ /kg	u kJ/kg	h kJ/kg	s kJ/kg.K
8000	(295.04)	(0.02352)	(2569.6)	(2757.8)	(5.7431)
	300	0.02426	2590.5	2784.6	5.7901
	350	0.02995	2746.7	2986.3	6.1286
	400	0.03431	2863.5	3138.0	6.3630
	450	0.03816	2966.9	3272.2	6.5554
	500	0.04174	3064.6	3398.5	6.7243
	550	0.04515	3159.8	3521.0	6.8778
	600	0.04845	3254.0	3641.5	7.0200
	650	0.05166	3348.1	3761.4	7.1535
	700	0.05482	3442.8	3881.4	7.2800
	750	0.05793	3538.3	4001.7	7.4007
	800	0.06101	3634.9	4122.9	7.5163
	850	0.06406	3732.6	4245.1	7.6275

P kPa	T °C	v m ³ /kg	u kJ/kg	h kJ/kg	s kJ/kg.K
15,000	(342.19)	(0.01034)	(2455.0)	(2610.1)	(5.3092)
	350	0.01147	2519.3	2691.3	5.4404
	400	0.01565	2739.9	2974.7	5.8799
	450	0.01845	2879.9	3156.6	6.1410
	500	0.02080	2997.3	3309.3	6.3452
	550	0.02292	3104.9	3448.8	6.5201
	600	0.02490	3207.9	3581.5	6.6767
	650	0.02679	3308.6	3710.5	6.8204
	700	0.02862	3408.3	3837.6	6.9544
	750	0.03039	3507.8	3963.7	7.0808
	800	0.03213	3607.6	4089.6	7.2009
	850	0.03384	3708.1	4215.6	7.3158

P kPa	T °C	v m ³ /kg	u kJ/kg	h kJ/kg	s kJ/kg.K
25,000	500	0.01112	2886.1	3164.2	5.9616
	550	0.01272	3018.6	3336.5	6.1778
	600	0.01413	3137.3	3490.4	6.3593
	650	0.01542	3249.0	3634.5	6.5198
	700	0.01664	3356.8	3773.0	6.6659
	750	0.01781	3462.7	3908.0	6.8012
	800	0.01894	3567.6	4041.1	6.9282
	850	0.02003	3672.3	4173.1	7.0485

P kPa	T °C	v m ³ /kg	u kJ/kg	h kJ/kg	s kJ/kg.K
35,000	600	0.009511	3061.9	3394.7	6.1174
	650	0.01056	3186.5	3556.0	6.2971
	700	0.01152	3303.6	3706.9	6.4563
	750	0.01243	3416.5	3851.6	6.6012
	800	0.01330	3526.9	3992.2	6.7355
	850	0.01413	3636.0	4130.4	6.8614

P kPa	T °C	v m ³ /kg	u kJ/kg	h kJ/kg	s kJ/kg.K
10,000	(311.03)	(0.01803)	(2544.2)	(2724.5)	(5.6139)
	350	0.02242	2698.1	2922.2	5.9425
	400	0.02641	2832.0	3096.1	6.2114
	450	0.02975	2943.6	3241.1	6.4194
	500	0.03278	3046.2	3374.0	6.5971
	550	0.03563	3144.6	3500.9	6.7561
	600	0.03836	3241.1	3624.7	6.9022
	650	0.04101	3337.1	3747.1	7.0385
	700	0.04359	3433.1	3869.0	7.1671
	750	0.04613	3529.7	3991.0	7.2893
	800	0.04863	3627.2	4113.5	7.4062
	850	0.05110	3725.7	4236.7	7.5184

P kPa	T °C	v m ³ /kg	u kJ/kg	h kJ/kg	s kJ/kg.K
20,000	(365.80)	(0.005874)	(2296.1)	(2413.6)	(4.9330)
	400	0.009946	2617.9	2816.9	5.5521
	450	0.01270	2806.8	3060.8	5.9026
	500	0.01477	2944.1	3239.4	6.1417
	550	0.01655	3063.0	3393.9	6.3355
	600	0.01817	3173.3	3536.7	6.5039
	650	0.01969	3279.2	3672.9	6.6557
	700	0.02113	3382.8	3805.5	6.7955
	750	0.02253	3485.4	3936.0	6.9263
	800	0.02388	3587.8	4065.4	7.0498
	850	0.02521	3690.3	4194.4	7.1673

P kPa	T °C	v m ³ /kg	u kJ/kg	h kJ/kg	s kJ/kg.K
30,000	500	0.008676	2823.2	3083.5	5.7936
	550	0.01016	2972.0	3276.8	6.0362
	600	0.01143	3100.1	3443.1	6.2324
	650	0.01258	3218.0	3595.5	6.4022
	700	0.01365	3330.4	3740.1	6.5547
	750	0.01467	3439.7	3879.8	6.6948
	800	0.01564	3547.3	4016.7	6.8254
	850	0.01659	3654.2	4151.8	6.9484

P kPa	T °C	v m ³ /kg	u kJ/kg	h kJ/kg	s kJ/kg.K
40,000	650	0.009046	3154.5	3516.3	6.2012
	700	0.009930	3276.6	3673.8	6.3673
	750	0.01075	3393.1	3823.3	6.5171
	800	0.01154	3506.4	3967.8	6.6551
	850	0.01229	3617.7	4109.2	6.7838